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Kim et al.

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(54) **HIGH CAPACITY MEMORY CIRCUIT WITH LOW EFFECTIVE LATENCY**

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G06F 3/06 (2006.01)
(Continued)

(52) **U.S. Cl.**
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(58) **Field of Classification Search**
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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

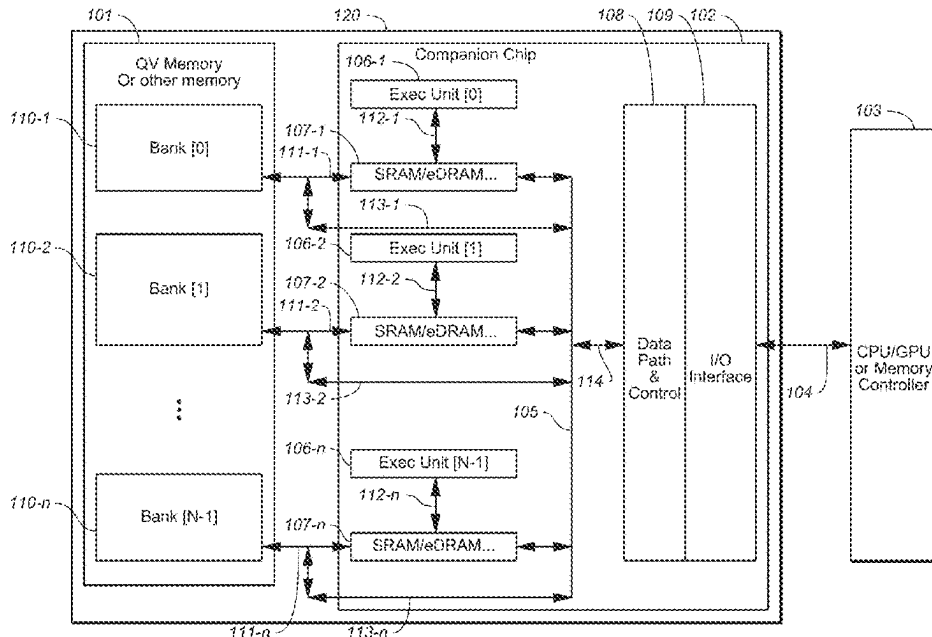
11,675,500 B2 * 6/2023 Kim G06F 3/0631 711/118
11,910,612 B2 2/2024 Yan et al.
(Continued)

Primary Examiner — Pho M Luu

(57) **ABSTRACT**

A first circuit formed on a first semiconductor substrate is wafer-bonded to a second circuit formed on a second memory circuit, wherein the first circuit includes quasi-volatile or non-volatile memory circuits and wherein the second memory circuit includes fast memory circuits that have lower read latencies than the quasi-volatile or non-volatile memory circuits, as well as logic circuits. The volatile and non-volatile memory circuits may include static random-access memory (SRAM) circuits, dynamic random-access memory (DRAM) circuits, embedded DRAM (eDRAM) circuits, magnetic random-access memory (MRAM) circuits, embedded MRAM (eMRAM), or any suitable combination of these circuits.

73 Claims, 24 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/169,387, filed on
Feb. 5, 2021, now Pat. No. 11,675,500.

(60) Provisional application No. 62/971,720, filed on Feb.
7, 2020.

(51) **Int. Cl.**

H01L 23/00 (2006.01)

H01L 25/18 (2023.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

12,073,082	B2 *	8/2024	Kim	G06F 3/0611
2018/0364944	A1	12/2018	Grossman et al.	
2019/0198082	A1	6/2019	O	
2022/0083254	A1 *	3/2022	Lee	G11C 13/0061
2024/0403235	A1 *	12/2024	Ali	G06F 13/4027

* cited by examiner

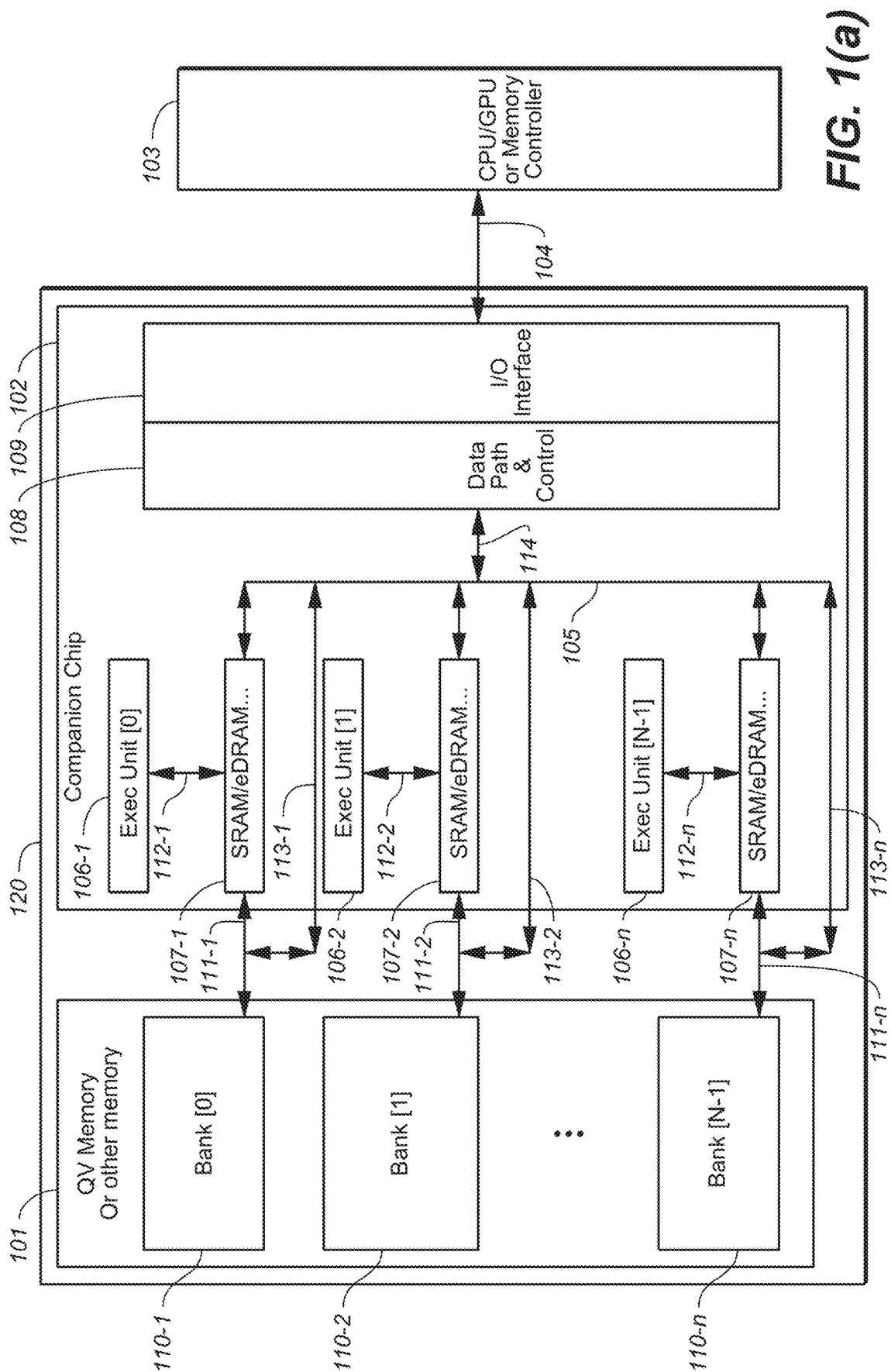


FIG. 1(a)

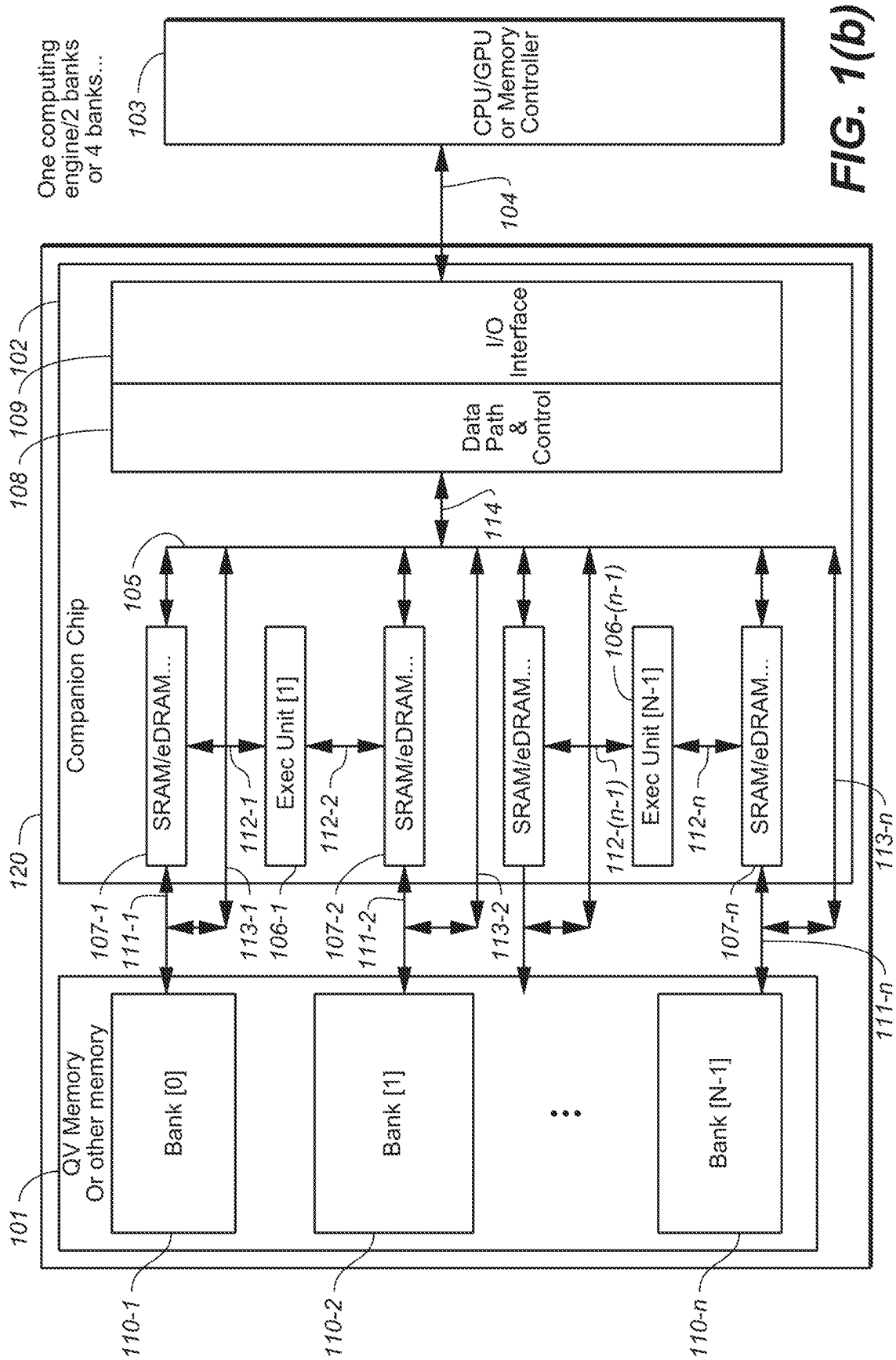
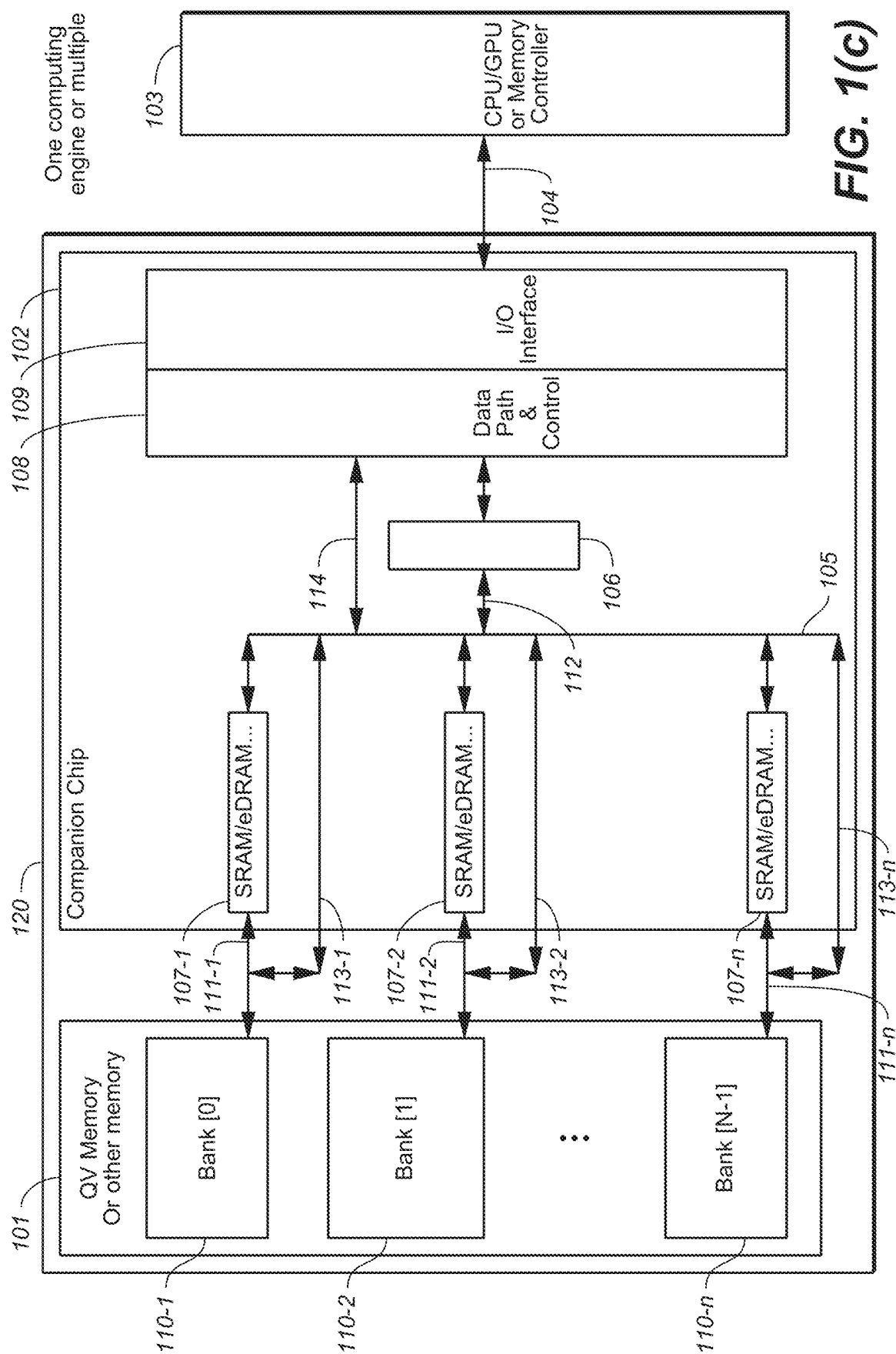


FIG. 1(b)



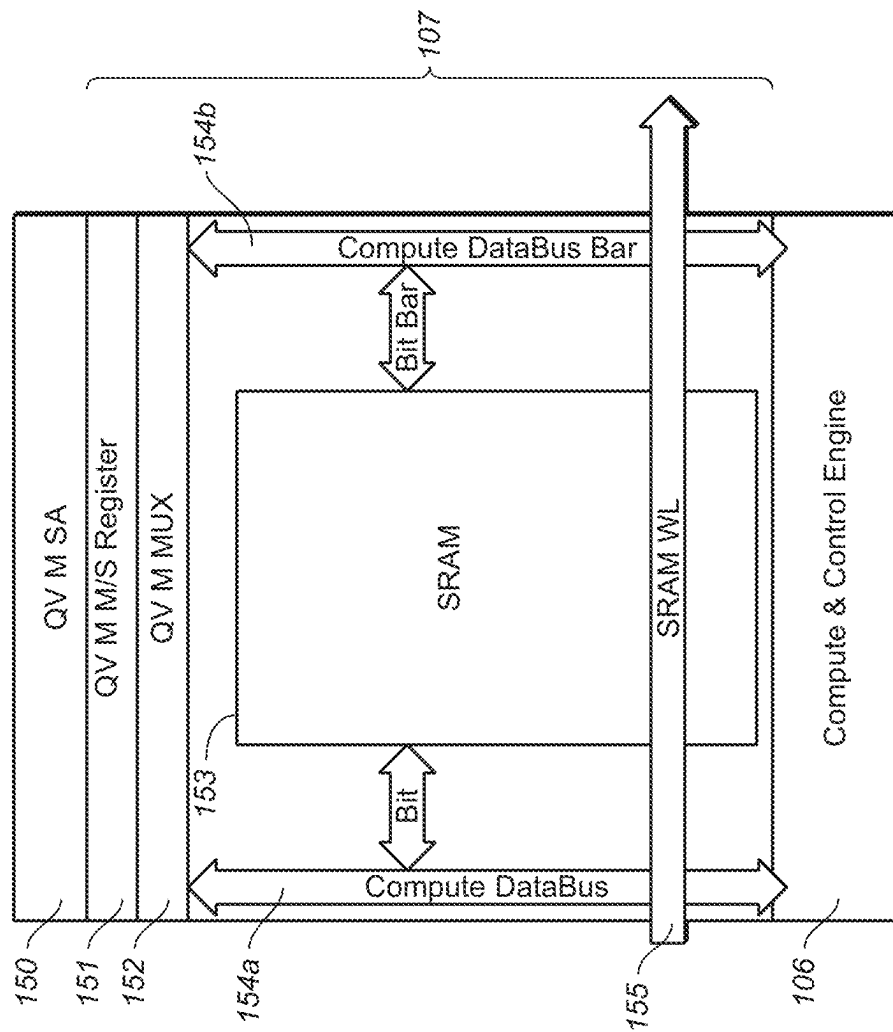


FIG. 1(d)

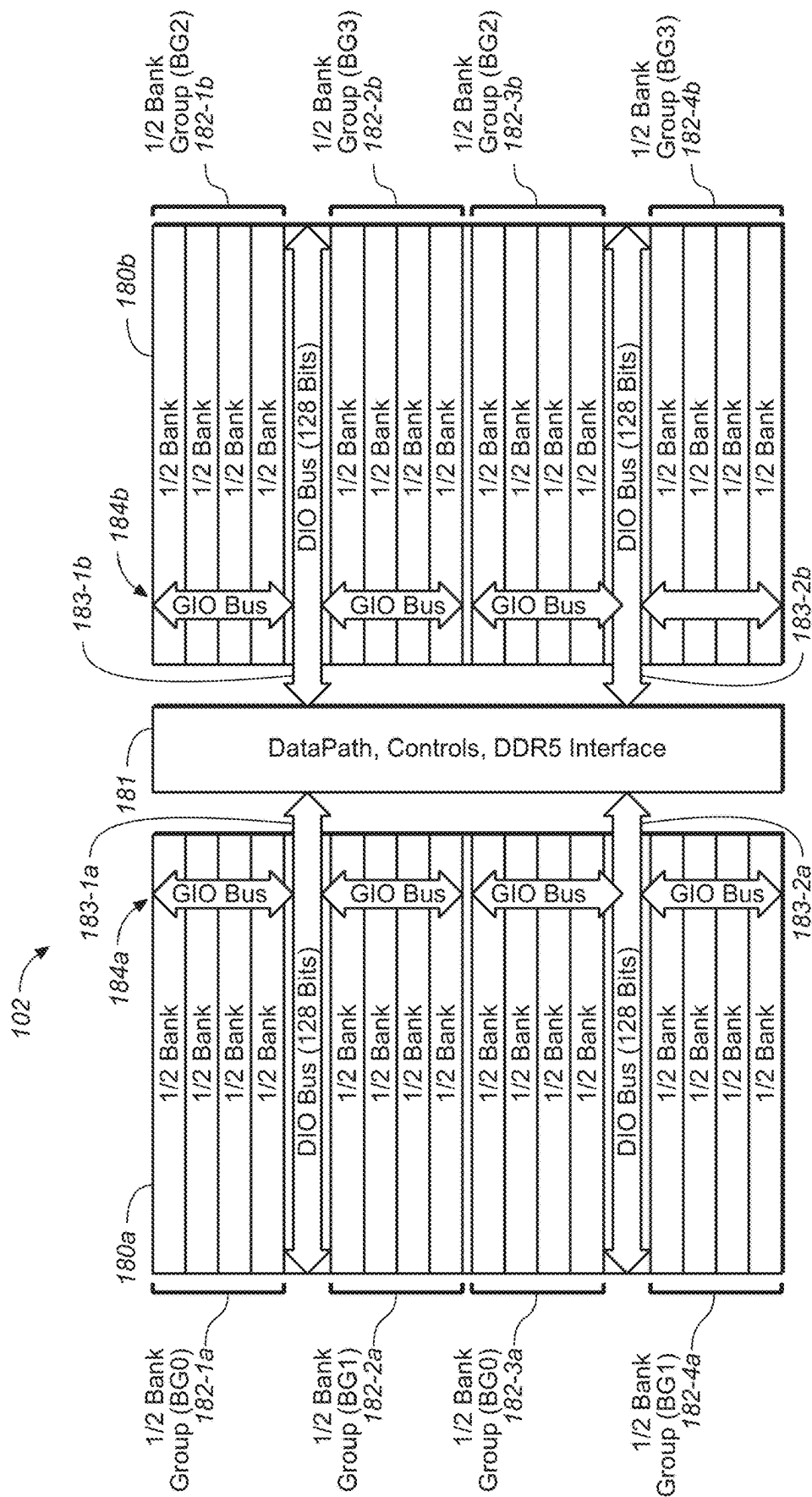


FIG. 1(e)

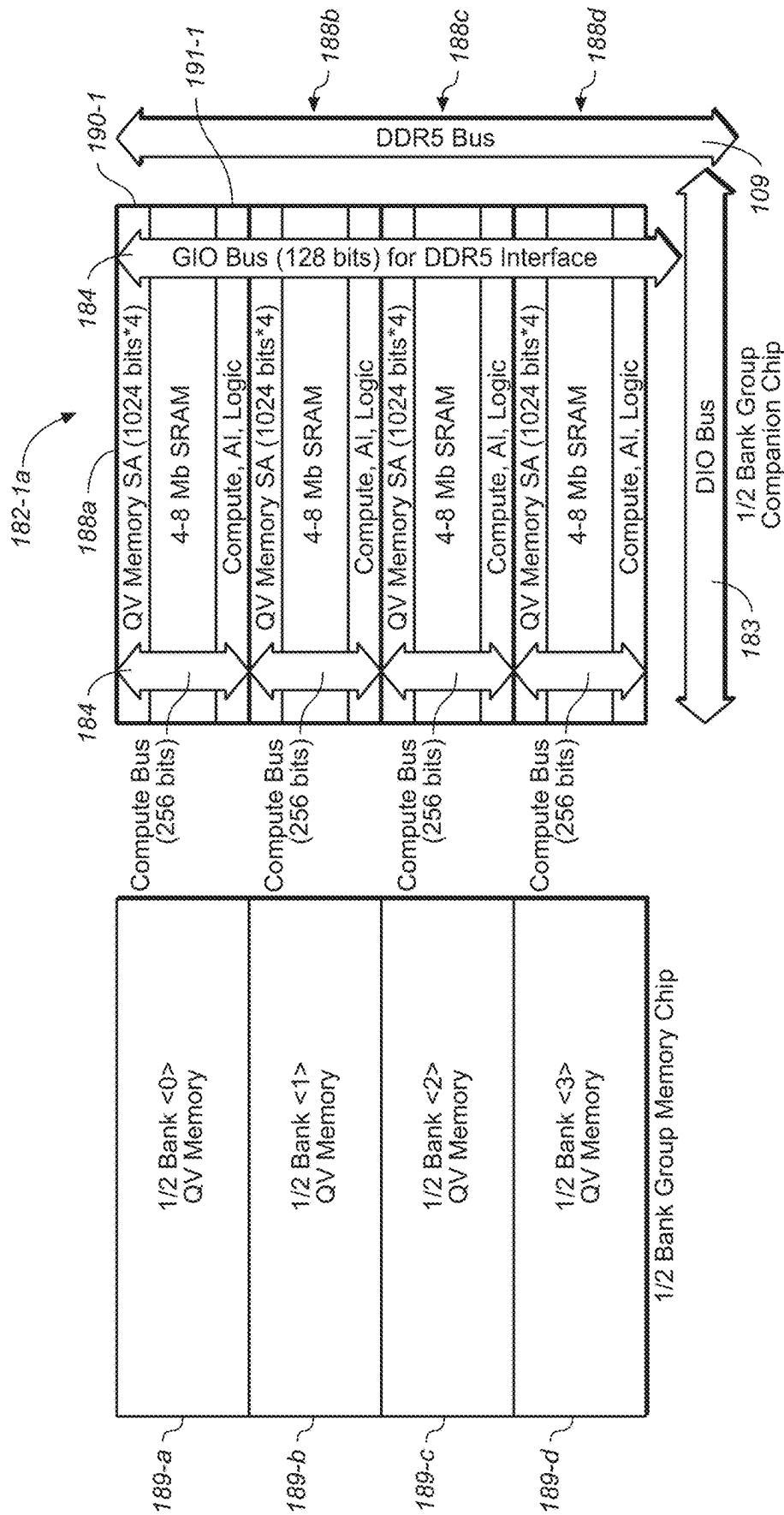


FIG. 1(f)

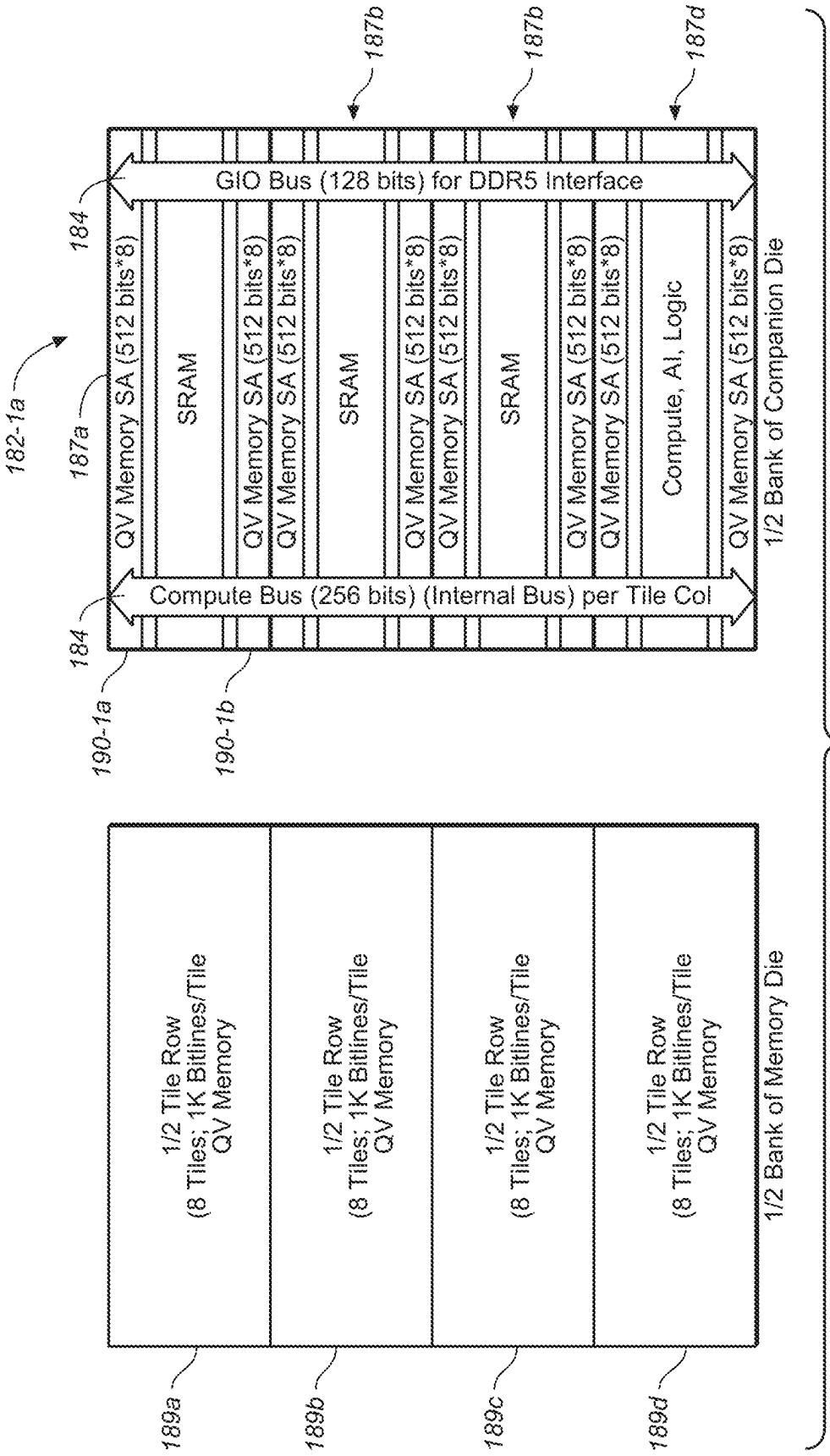


FIG. 1(g)

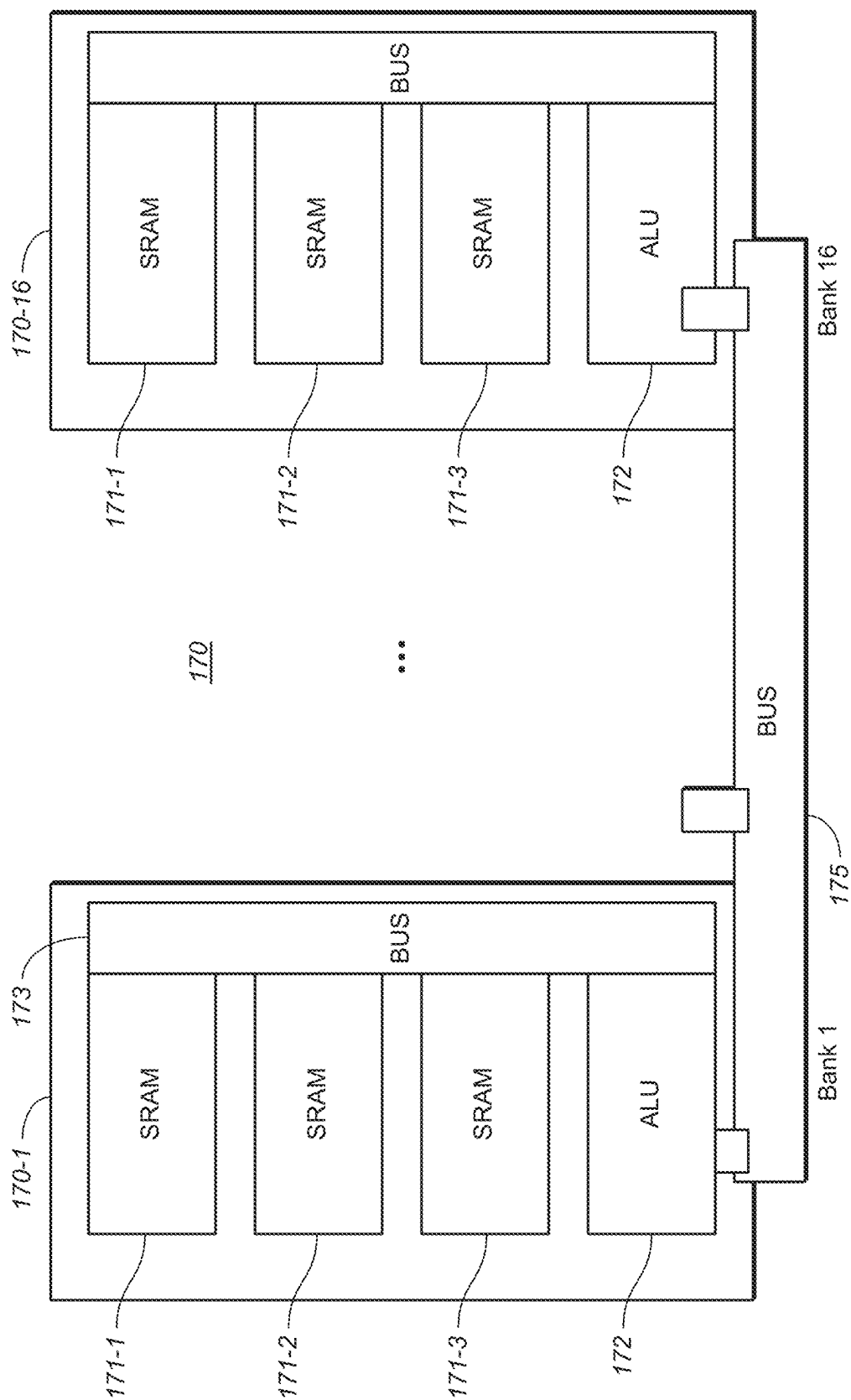


FIG. 1(h)

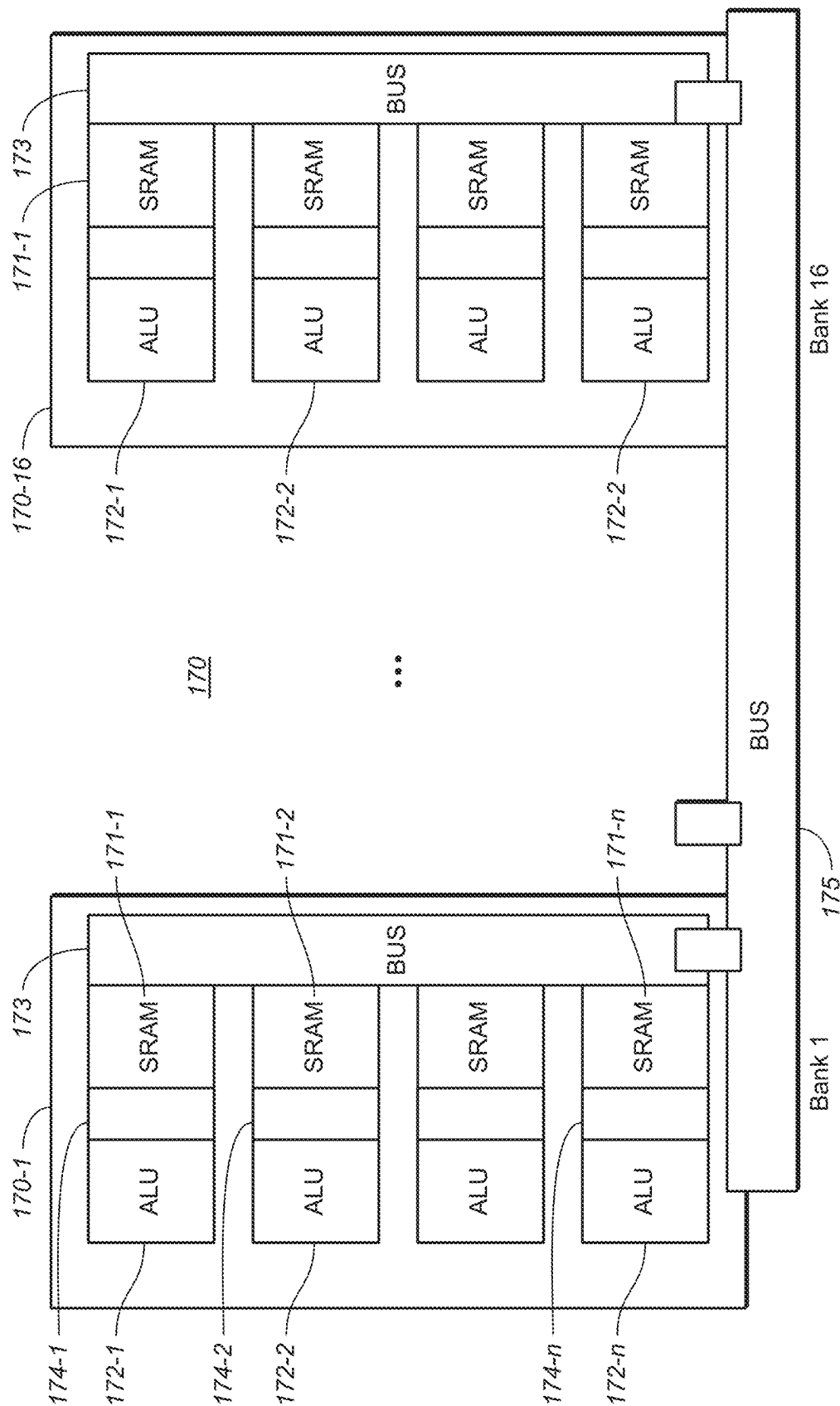


FIG. 1(i)

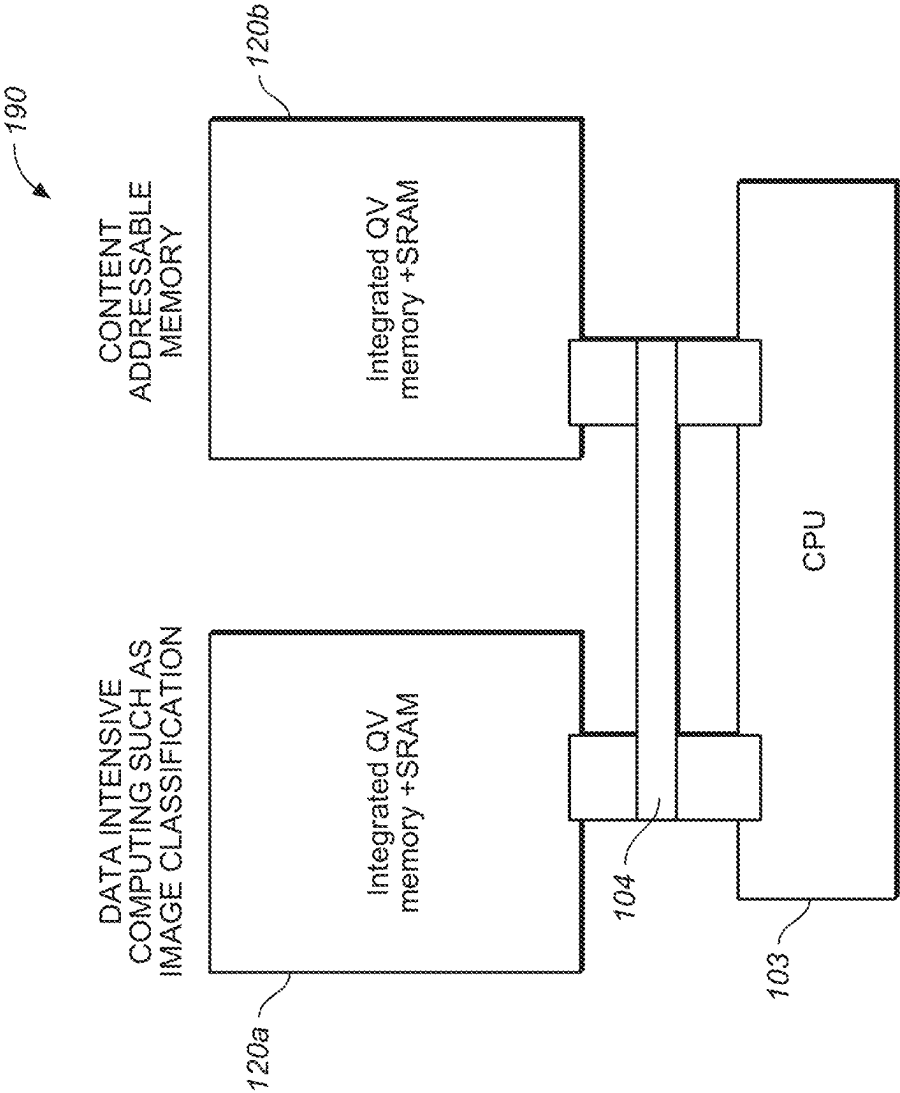


FIG. 1(j)

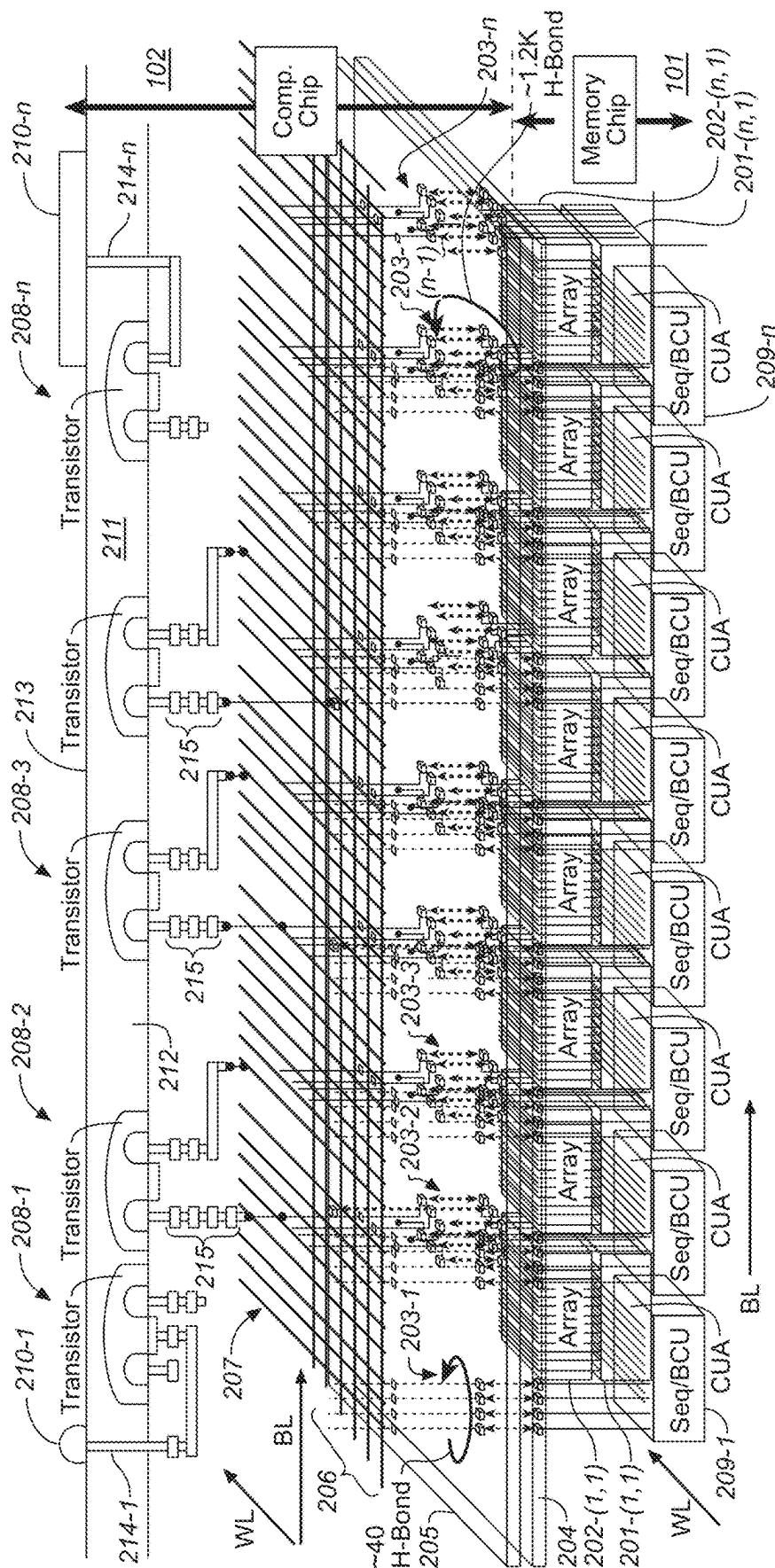


FIG. 2(a)

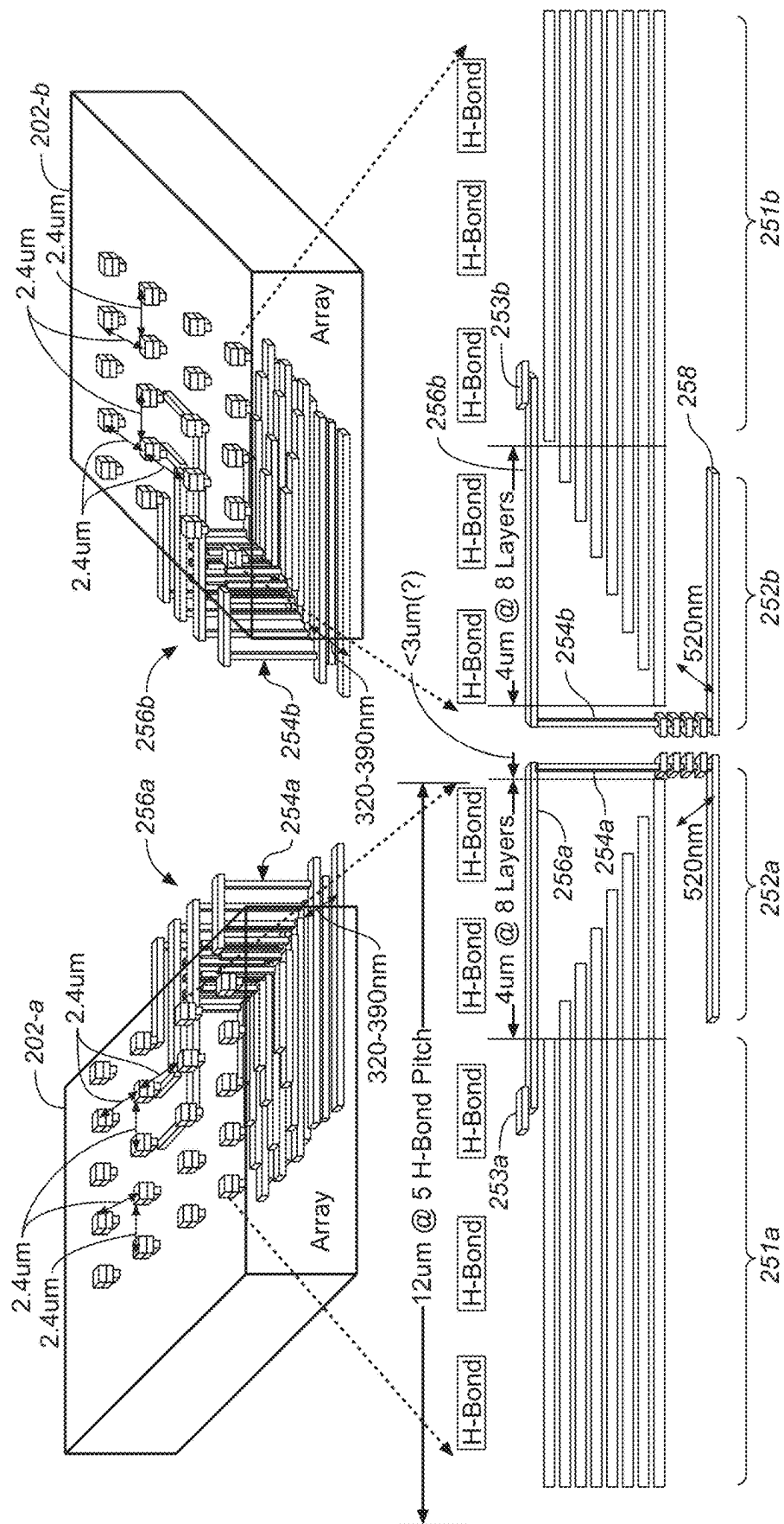


FIG. 2(b)

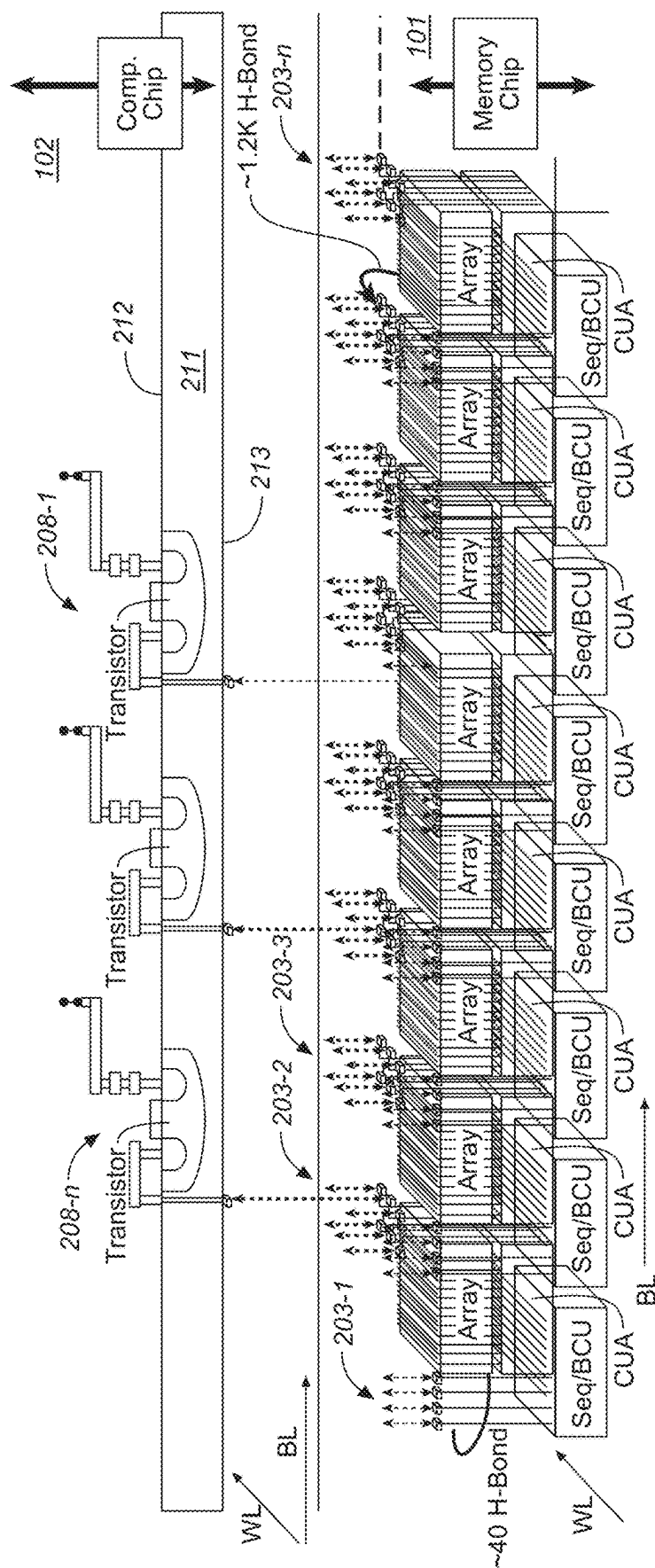


FIG. 2(c)

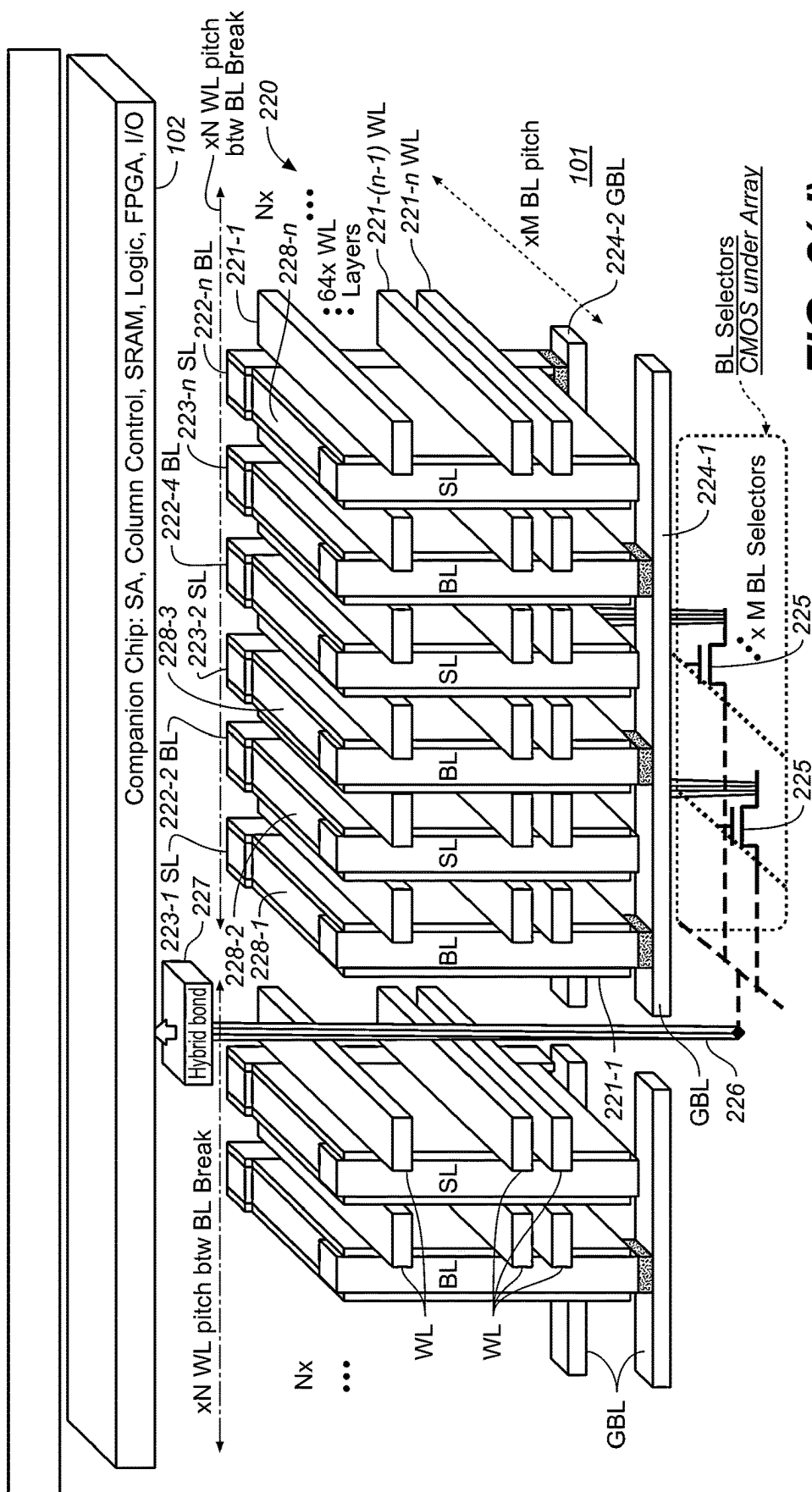


FIG. 2(d)

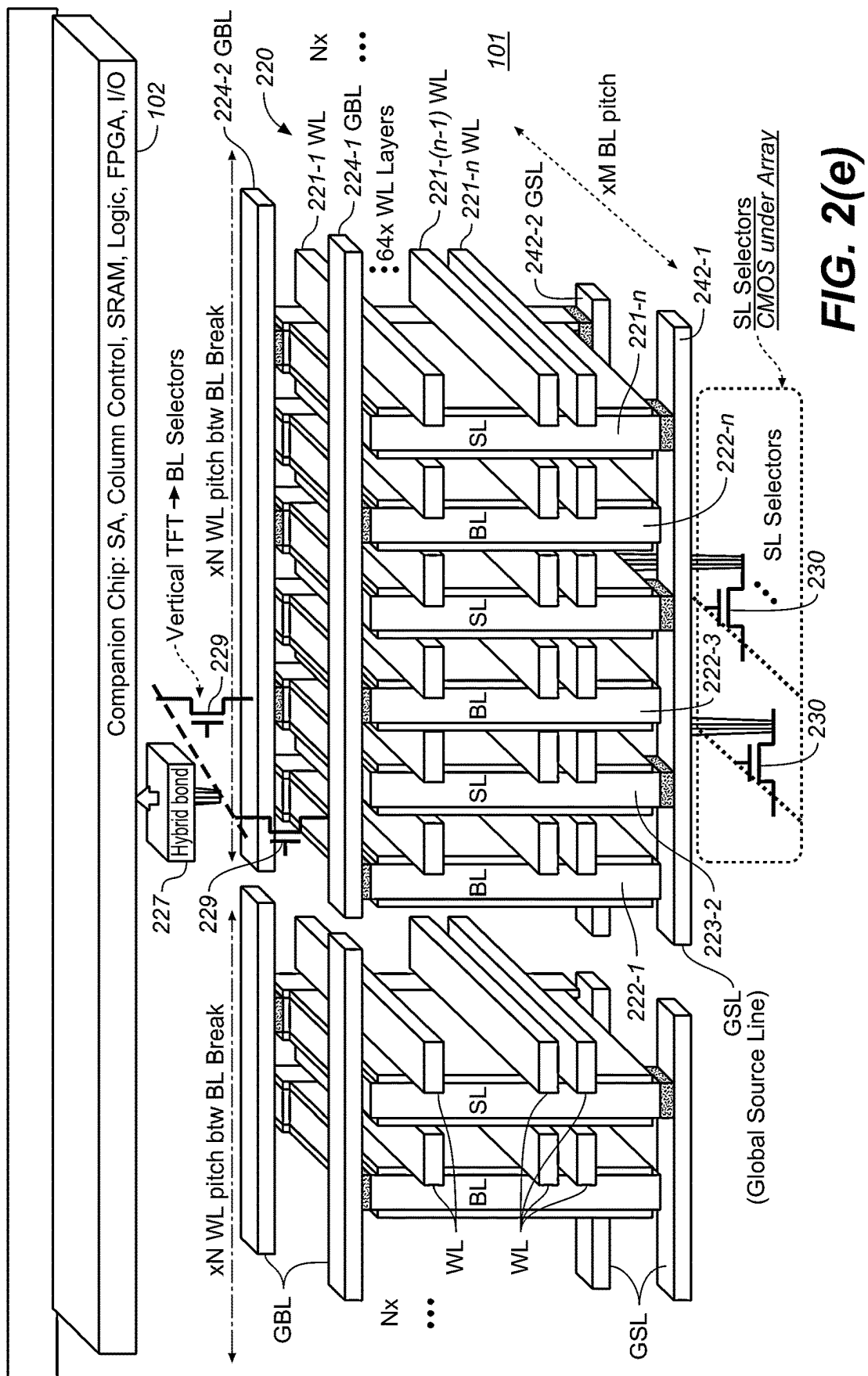
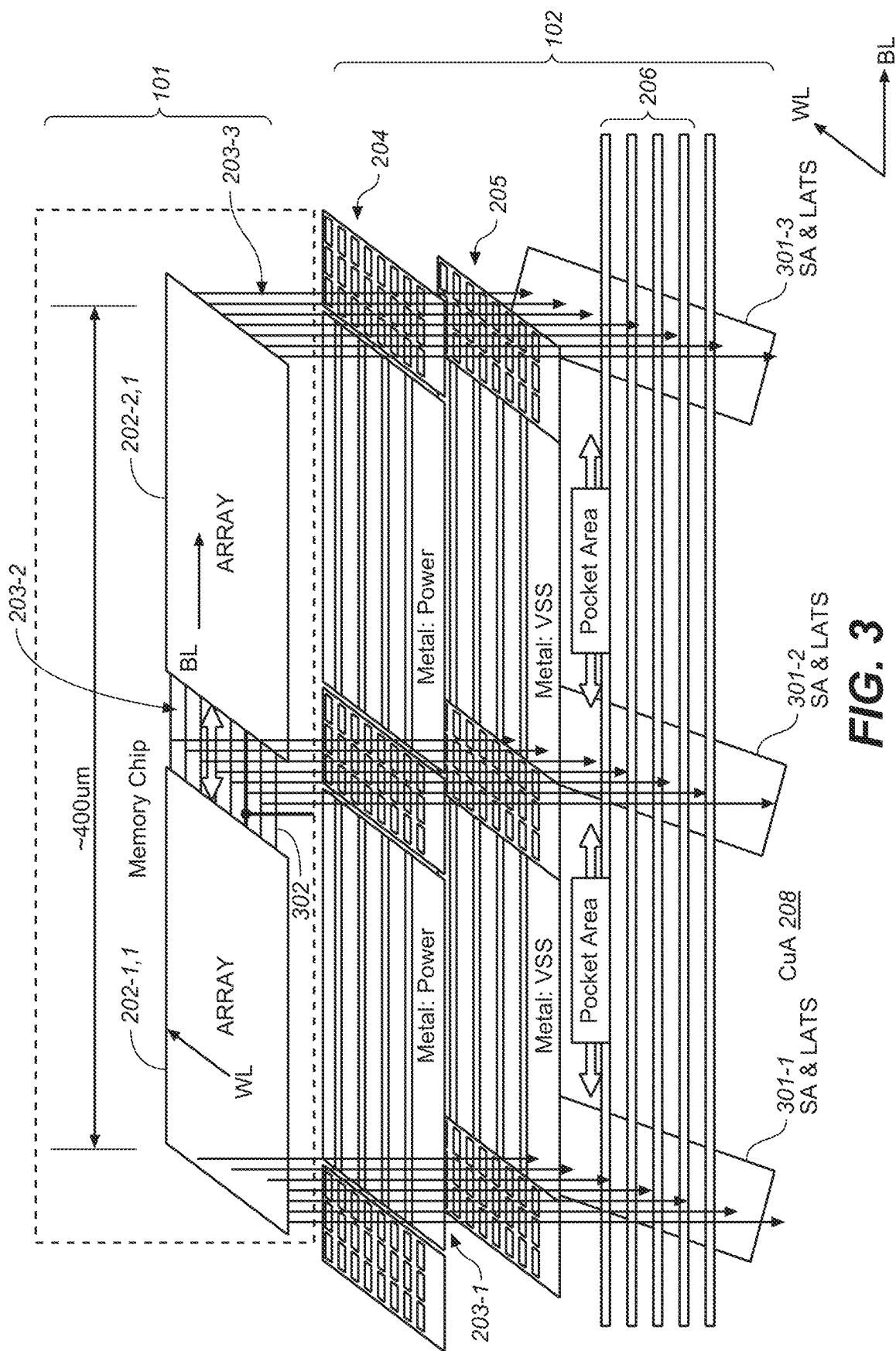


FIG. 2(e)



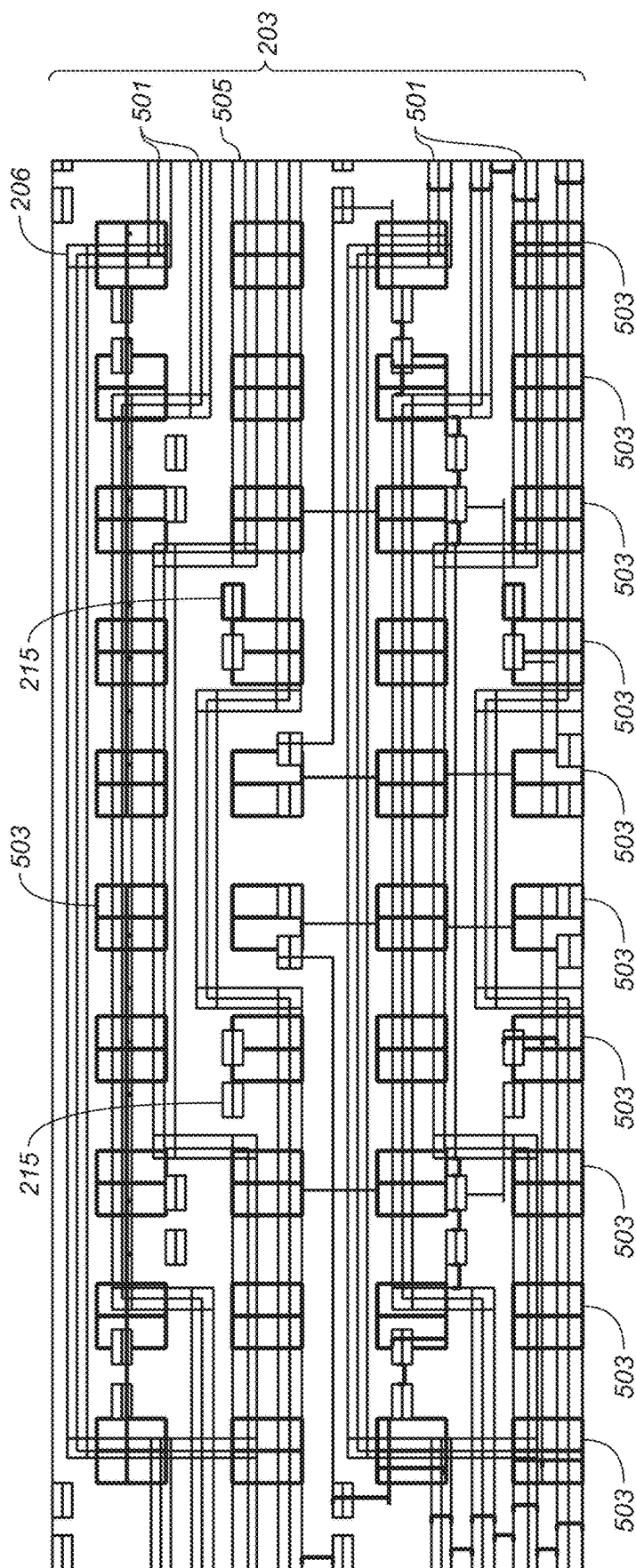
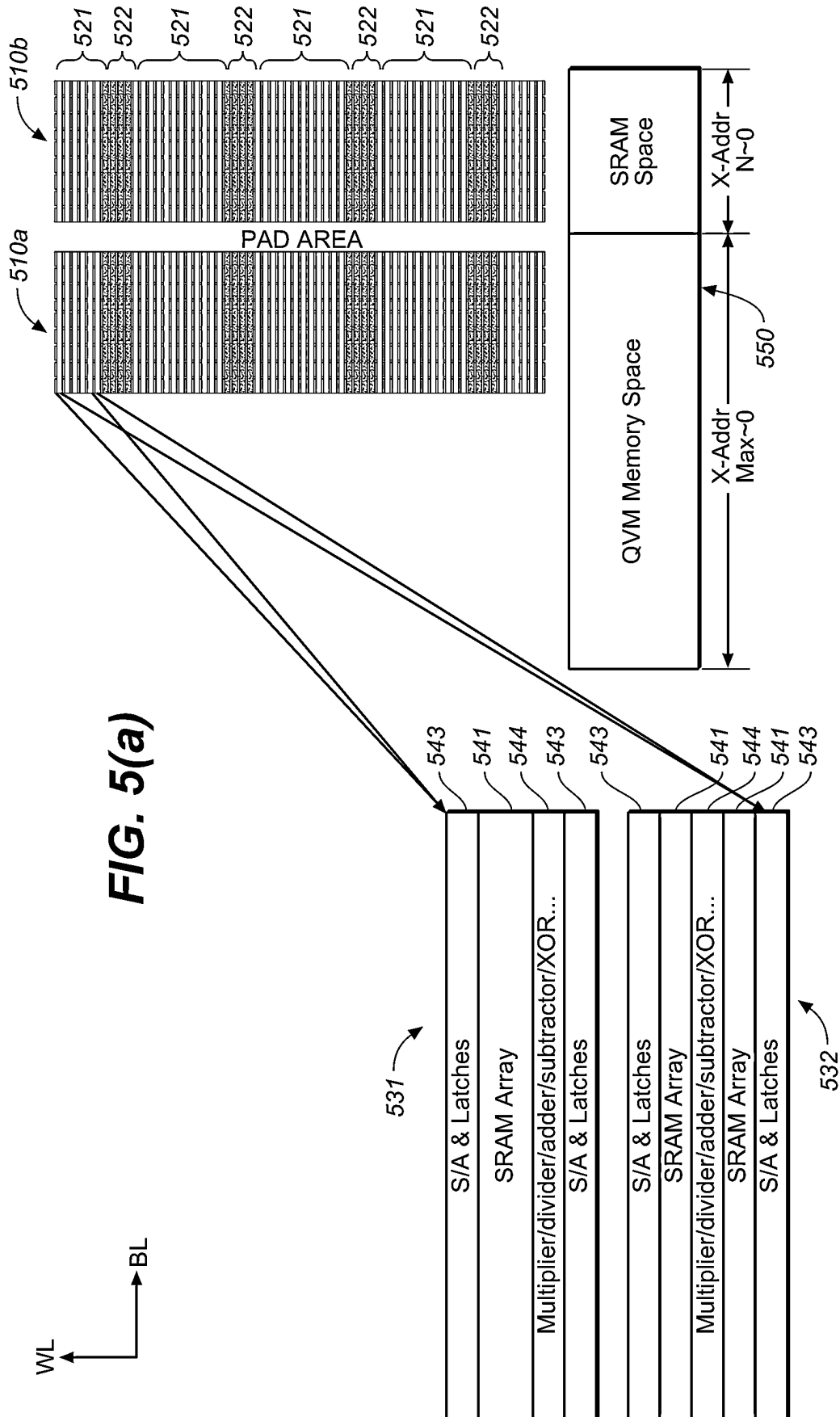
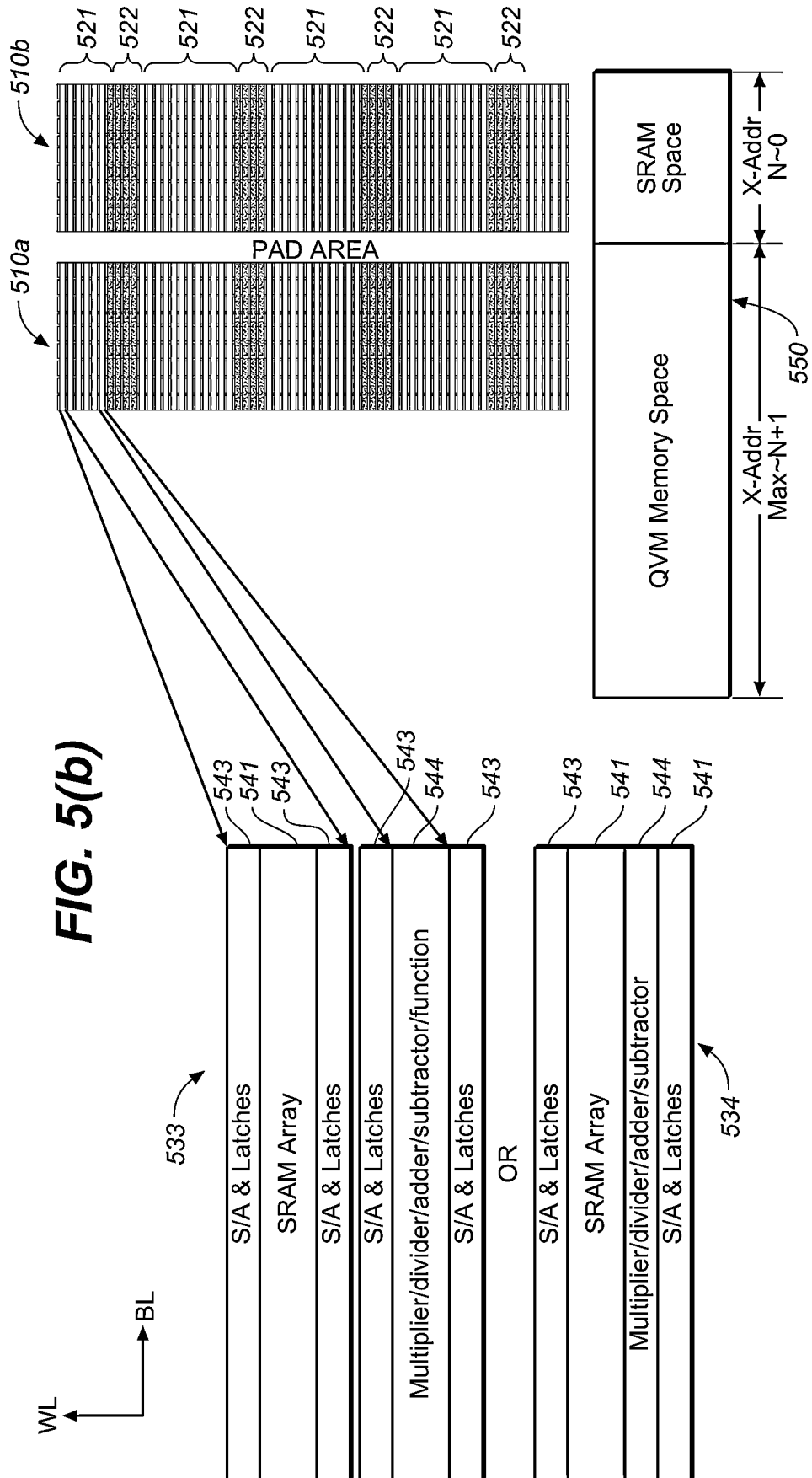


FIG. 4





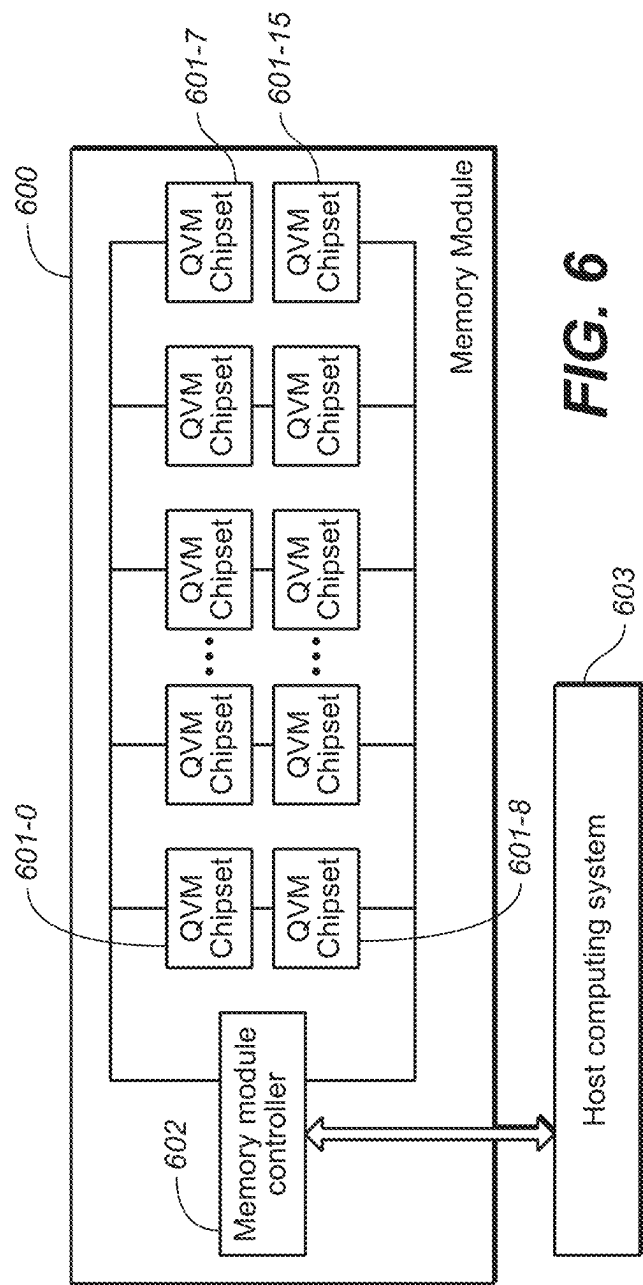
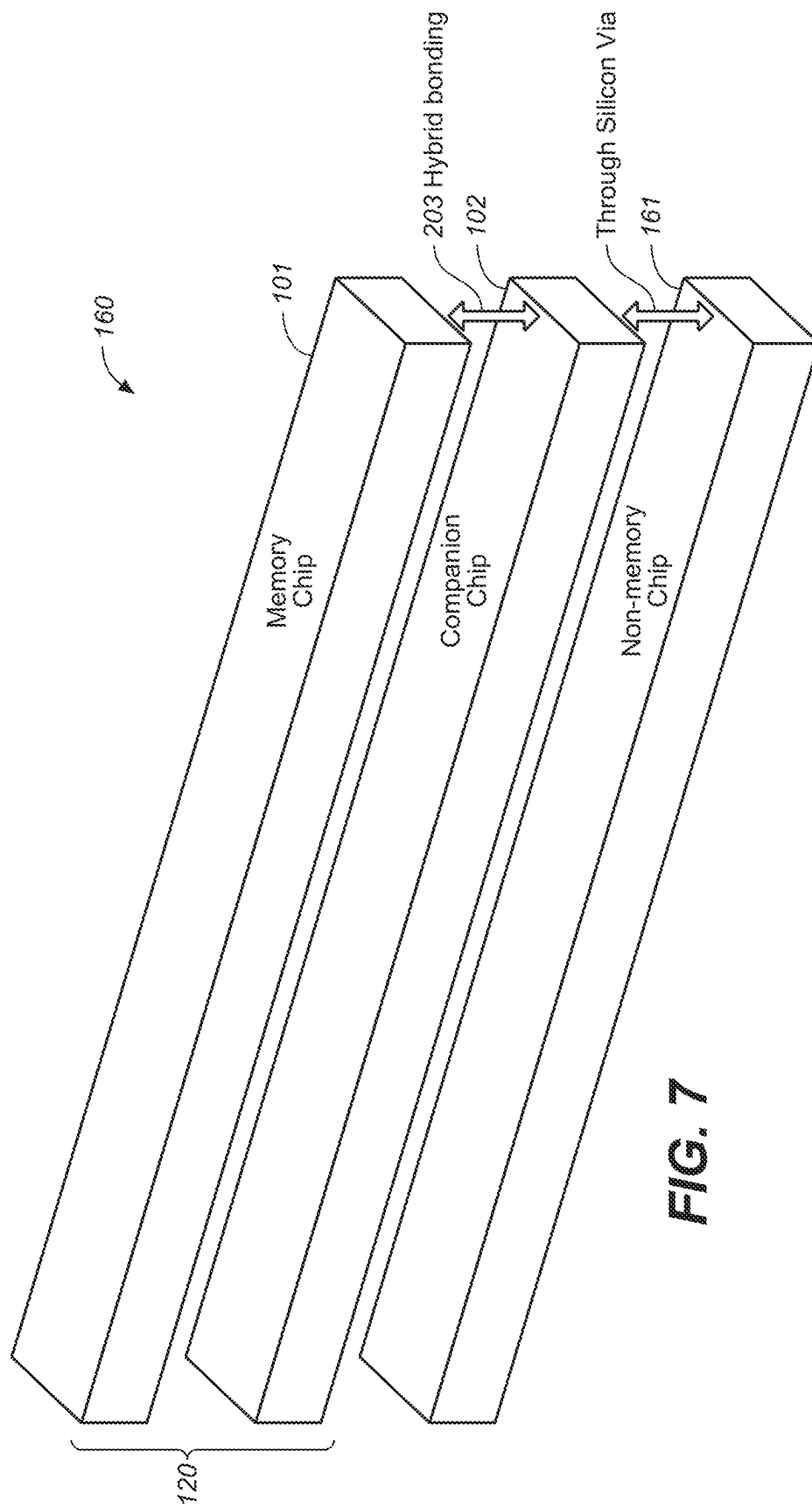
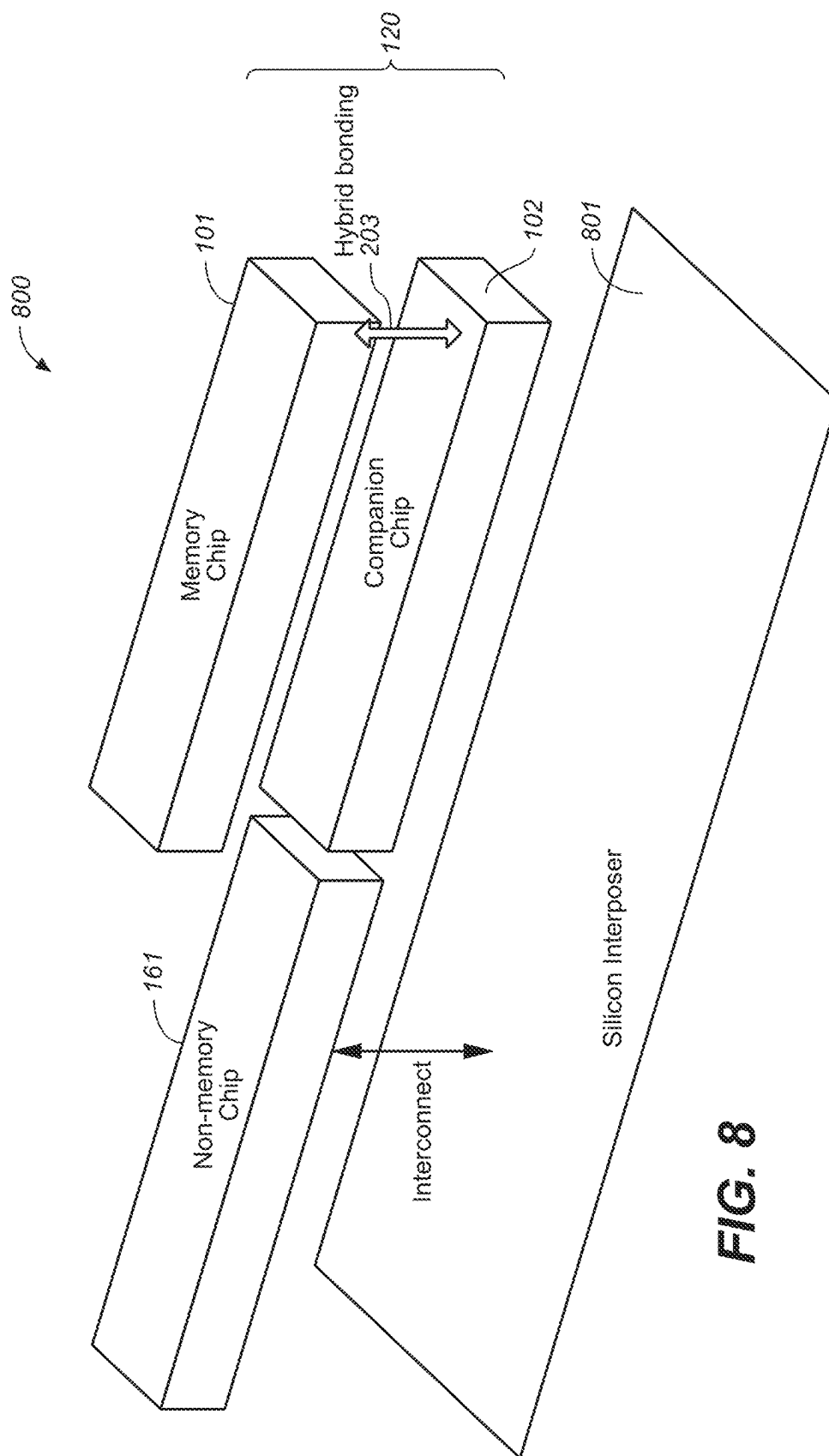
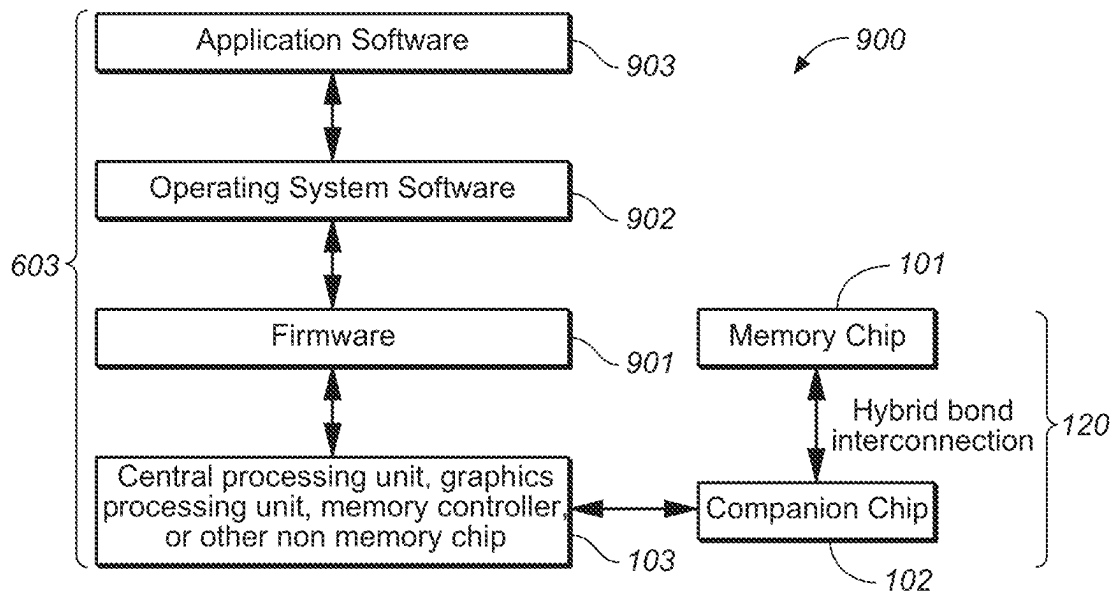
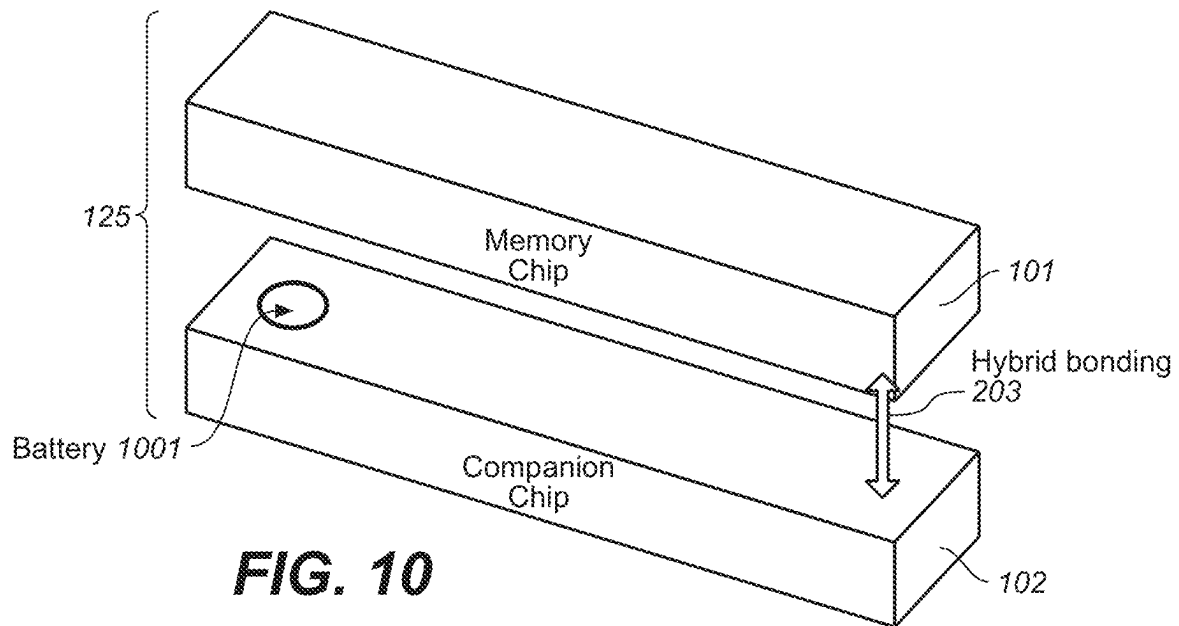


FIG. 6





**FIG. 9****FIG. 10**

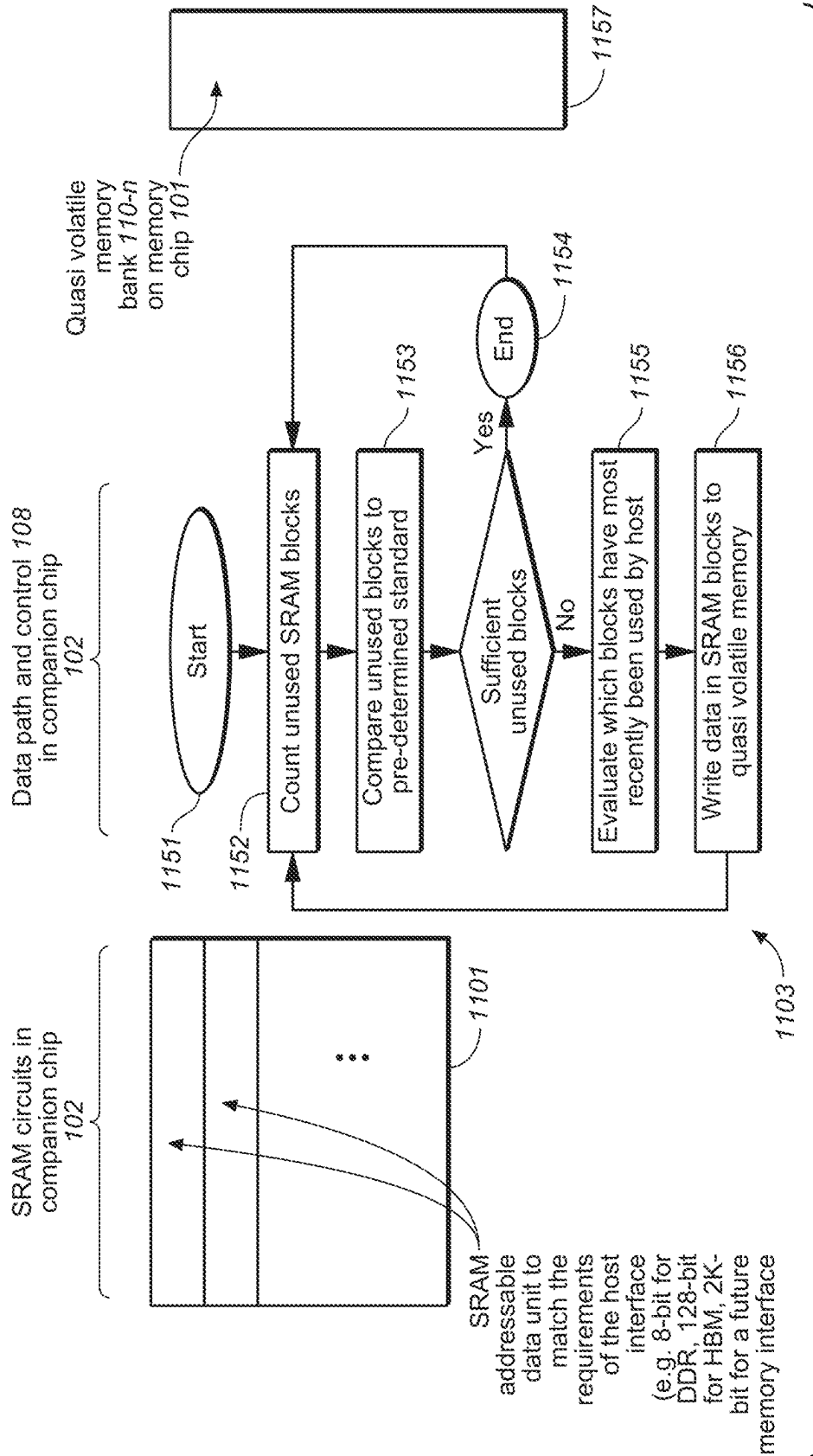


FIG. 11

HIGH CAPACITY MEMORY CIRCUIT WITH LOW EFFECTIVE LATENCY

CROSS REFERENCE TO RELATED APPLICATION

The present application is related to and claims priority of U.S. patent application ("Parent application"), Ser. No. 18/306,073, entitled "High Capacity Memory Circuit With Low Effective Latency," filed on Apr. 24, 2023, now U.S. Pat. No. 12,073,082, which is a continuation application of U.S. patent application Ser. No. 17/169,387, entitled "High Capacity Memory Circuit With Low Effective Latency," filed on Feb. 5, 2021, now U.S. Pat. No. 11,675,500, which claims priority to provisional patent application ("Parent Provisional Application"), Ser. No. 62/971,720, entitled "High Capacity Memory Circuit With Low Effective Latency," filed on Feb. 7, 2020.

The present application is also related to (i) U.S. non-provisional application ("Non-provisional Application I"), Ser. No. 16/776,279, entitled "Device with Embedded High-Bandwidth, High-Capacity Memory using Wafer Bonding," filed on Jan. 29, 2020, and (ii) U.S. patent application ("Non-provisional Application II"), Ser. No. 16/582,996, entitled "Memory Circuit, System and Method for Rapid Retrieval of Data Sets," filed on Sep. 25, 2019; (iii) U.S. non-provisional patent application ("Non-provisional Application III"), Ser. No. 16/593,642, entitled "Three-dimensional Vertical NOR Flash Thin-film Transistor Strings," filed on Oct. 4, 2019; and (iv) U.S. non-provisional patent application ("Non-provisional application IV"), Ser. No. 16/744,067, entitled "Implementing Logic Function and Generating Analog Signals Using NOR Memory Strings," filed on Jan. 15, 2020.

The present application is also related to U.S. provisional application ("Provisional Application"), Ser. No. 62/947,405, entitled "Vertical Thin-film Transistor and Application as Bit Line Connector for 3-Dimensional Memory Arrays," filed on Dec. 12, 2019.

The disclosures of Parent Application, Parent Provisional Application, Provisional Application and Non-provisional Applications I-IV are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to memory circuits and computing systems. In particular, the present invention relates to memory circuits that are very high capacity, while providing a low effective latency comparable to state-of-the-art dynamic random-access memory ("DRAM") circuits and the interactions between memory and computer systems.

2. Discussion of the Related Art

Non-Provisional Applications II and III each disclose high-capacity, 3-dimensional thin-film memory circuits that can be configured as quasi-volatile memory circuits. A quasi-volatile memory circuit, though having a shorter data retention time (e.g., minutes) compared to the data retention time of a non-volatile memory circuit (e.g., years), has faster write and erase operations, greater endurance and lower read latency than conventional non-volatile circuits as well as comparable circuit density. Non-provisional Applications II

and III also each disclose forming the quasi-volatile memory circuits as 3-dimensional arrays of thin-film storage transistors over a semiconductor substrate in which is formed analog and digital support circuitry, such as various power supply circuits, drivers, sense amplifiers, word line and bit line decoding circuits, data latches, multiplexers, select transistors and input and output circuits. Some of these circuits may operate at high voltage (e.g., 8.0-16.0 volts), while others operate at medium-voltage (e.g., 2.0-6.0 volts) and low voltages (e.g., 0.6-1.2 volts). In this description, the circuitry formed in the semiconductor substrate underneath the 3-dimensional memory arrays of thin-film storage transistors are generally referred to as "circuitry under array" ("CuA"). Typically, for non-volatile or quasi volatile thin-film memory arrays, the high-voltage circuits are relatively low-density (i.e., large area) circuits, while low-voltage transistors are relatively high density. Among these transistor types, the low-voltage transistors typically have the highest performance (i.e., fastest) and provide densest circuits.

In one disclosed embodiment in Non-provisional Application II, the storage transistors of each 3-dimensional array are organized into parallel stacks of NOR memory strings, with the stack having eight or more NOR memory strings provided one on top of another, separated by a dielectric layer. The storage transistors in each NOR memory string share a common drain region and a common source region. The common drain region of each NOR memory string, also colloquially referred to as a "bit line," extends along a direction parallel to the surface of the semiconductor substrate. Connections to the gate electrodes of the storage transistors are provided by conductors ("word lines") that are shared by numerous NOR memory strings. Each word line extends along a direction substantially perpendicular to the surface of the semiconductor substrate. In this detailed description, the memory arrays of Non-provisional Application II are referred to as HNOR memory arrays, based on their substantially "horizontal" common drain and common source regions.

As disclosed in Non-provisional Application II, the storage transistors in the 3-dimensional memory array form a storage portion ("array portion") and a contact portion ("staircase portion"). The staircase portion is so named because each bit line of each stack of NOR memory strings extends beyond the array portion a successively lesser amount, as the distance between the bit line and the surface of the semiconductor substrate increases, so as to form a staircase structure. Electrical contacts to the bit lines may be provided at the staircase portion. The staircase portion of each stack of NOR memory strings may have two staircase structures on opposite sides of the array portion.

In one disclosed embodiment in Non-provisional Application III, the storage transistors of each 3-dimensional array are organized into parallel columns of NOR memory strings, with each column having at least one NOR memory string, in which storage transistors share a common drain region and a common source region. The common drain region or bit line of each NOR memory string extends along a direction substantially perpendicular the surface of the semiconductor substrate. In this detailed description, the memory arrays of Non-provisional Application III are referred to as VNOR memory arrays, based on their substantially "vertical" common drain and common source regions. Like the HNOR memory arrays, storage transistors in the 3-dimensional VNOR memory array also form a storage portion ("array portion") and a contact portion ("staircase portion"). The staircase portion of a VNOR memory array provides

electrical contacts to the word lines. Electrical contacts to the bit lines may be provided at the staircase portion. The staircase portion of a VNOR memory array may have two staircase structures on opposite sides of the array portion.

Forming thin-film memory arrays over the CuA poses challenges. For example, manufacturing the quasi-volatile and non-volatile memory arrays above the substrate requires high temperature steps (“thermal cycles”). As the CuA is formed first in the substrate, prior to the formation of the quasi-volatile and non-volatile memory arrays, the CuA is also exposed to the thermal cycles. The dense low-voltage logic circuit are particularly susceptible to degradation resulting from exposure to the thermal cycles. For example, sense amplifiers are particularly susceptible to degradation under thermal processing, which adversely impacts their sensitivity and signal integrity. Therefore, the CuA imposes limits on the thermal budget allowable for forming the memory arrays, so as to prevent the thermal cycles from degrading the performance of the high-performance, low-voltage and other types of transistors in the CuA. High-voltage and medium-voltage circuits, generally speaking, can withstand the thermal cycles without experience any significant adverse effects.

The large number of manufacturing steps required to form both the CuA and the memory circuits adversely affects the potential yield and performance. Non-provisional application I discloses an integrated circuit formed by wafer-level hybrid bonding of semiconductor dies. Using wafer-level or chip-level hybrid bonding, a memory circuit and its related CuA (“memory chip”) and a logic circuit (“companion chip”) may be independently fabricated on separate semiconductor substrates and brought together by interconnecting through aligned hybrid bonds provided on their respective bonding surfaces. In this detailed description, the term “bond” or “bonding” may refer to any wafer-level bonding techniques, chip-level bonding, or any combination of wafer-level bonding and chip-level bonding (e.g., wafer-to-wafer hybrid bonding, chip-to-chip hybrid bonding and chip-to-wafer hybrid bonding). Non-provisional Application I shows that such a combination not only alleviates challenges in the fabrication steps, the combination may give rise to both higher performance in memory circuits and new applications of memory circuits not previously possible.

U.S. Patent Application Publication 2019/0057974, entitled “Hybrid Bonding Contact Structure Of Three-Dimensional Memory Device” (“Lu”) by Z. Lu et al, filed on Jul. 26, 2018, discloses a 3-dimensional (3-D) NAND memory device formed by bonding two semiconductor substrates. In Lu, a 3-D NAND memory array is fabricated above the planar surface a first substrate and “peripheral circuits” are fabricated on the second substrate. The two substrates are bonded using in a “flip-chip” fashion using hybrid bonds. Just below the bonding surface of each substrate, Lu teaches forming an interconnection structure, such that, when the two substrates are bonded, the hybrid bonds connect the two interconnection structures together to form an interconnection network that connects the peripheral circuits and the 3-D NAND memory array.

Lu discloses that the peripheral circuits formed on the second substrate includes “a page buffer, a decoder (e.g., a row decoder and a column decoder), a latch, a sense amplifier, a driver, a charge pump, a current or voltage reference, or any active or passive components of the circuits (e.g., high-voltage and low-voltage transistors, diodes, resistors, or capacitors). In some embodiments, the one or more peripheral circuits can be formed on second substrate 510 using complementary metal-oxide-semiconductor (CMOS)

technology (also known as a “CMOS chip”))” (Lu, at paragraph [0125]). Note that, page buffers, decoders and sense amplifiers are low-voltage logic circuits that can take best advantage of the best performance of the advanced manufacturing process nodes, as discussed above. Drivers, charge pumps, current or voltage references are often medium-voltage and high-voltage analog circuits that are required in a 3-D NAND memory circuit, for example, for generating programming, erase, read and inhibit voltages. The medium-voltage or high-voltage circuitry are generally not as scalable as the low-voltage circuitry, making them less cost-effective when manufactured under advanced manufacturing process nodes. In addition, a multi-oxide CMOS technology is required to accommodate both high-voltage and low-voltage transistors on the same chip. Such a process compromises the scaling and the performance in the low-voltage transistors that would otherwise be possible. Thus, by placing both high-voltage, medium-voltage, and low-voltage circuits on the second substrate, Lu’s peripheral circuits can only be manufactured on the second substrate using a manufacturing process that is capable of forming all of the low-voltage logic circuits and the medium-voltage and high-voltage analog circuitry, thus compromising both the high-voltage and low-voltage transistors. Lu’s approach prevents the low-voltage logic circuits from taking advantage of the better performance and circuit density in the more advanced manufacturing process nodes.

SUMMARY

According to one embodiment of the present invention, a first circuit formed on a first semiconductor substrate is bonded to a second circuit formed on a second semiconductor substrate, wherein the first circuit includes quasi-volatile or non-volatile memory circuits and wherein the second memory circuit includes faster memory circuits than the quasi-volatile or non-volatile memory circuits. Such faster memory circuits may be volatile or non-volatile memory circuits. The faster memory circuits may include static random-access memory (SRAM) circuits, dynamic random-access memory (DRAM) circuits, embedded DRAM (eDRAM) circuits, magnetic random-access memory (MRAM) circuits, embedded MRAM (eMRAM) circuits, spin-transfer torque MRAM (ST-MRAM) circuits, phase-change memory (PCM), resistive random-access memory (RRAM), conductive bridging random-access memory (CBRAM), ferro-electric resistive random-access memory (FRAM), carbon nanotube memory, or any suitable combination of these circuits. Bonding the first and the second circuits may be accomplished using conventional techniques, such as wafer-level or chip-level hybrid bonding.

The integrated circuit of the present invention make possible many new applications because of high data density, high endurance and high-speed access achievable by the quasi-volatile memory circuit on the memory chip, while the faster memory circuits on the companion chip provide even faster access times, the combination resulting effectively in a high-density, low-latency memory circuit, essentially a heterogeneous memory with advantages that can be exploited in new applications. For example, the integrated circuit of the present invention is particularly suitable for in-memory computing or near-memory computing applications.

The present invention is better understood upon consideration of the detailed description below in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1(a) shows integrated circuit 120—which includes memory chip 101 and companion chip 102, interconnected by hybrid bonds—operating under control or supervision by host processor 103, according to one embodiment of the present invention.

FIG. 1(b) illustrates an alternative implementation of integrated circuit 120, in which each of modularized logic circuits 106 is provided access to two of fast memory circuits 107, in accordance to one embodiment of the present invention.

FIG. 1(c) illustrates another alternative implementation of integrated circuit 120, in which single modularized logic circuit 106 is provided access to each of fast memory circuits 107, in accordance to one embodiment of the present invention.

FIG. 1(d) shows a functional representation of one of fast memory circuits 107, according to one embodiment of the present invention.

FIG. 1(e) shows a functional organization of memory bank groups BG(0)-BG(3) of the fast memory circuits in companion chip 101, according to one embodiment of present invention.

FIG. 1(f) illustrates four half-banks 188a-188d of a half-bank group (e.g., half-bank group 182-1a of FIG. 1(e)) on companion chip 102, relative corresponding half-banks 189a-189d of quasi-volatile memory circuits in memory chip 101, according to one embodiment of the present invention.

FIG. 1(g) illustrates four half-banks 187a-187d of a half-bank group (e.g., half-bank group 182-1a of FIG. 1(e)) on companion chip 102, relative corresponding half-banks 189a-189d of quasi-volatile memory circuits in memory chip 101, according to another embodiment of the present invention.

FIG. 1(h) illustrates functional configuration of a 16-bank computing platform 170 based on organization such as that described above in conjunction with FIG. 1(e), according to one embodiment of the present invention.

FIG. 1(i) illustrates an alternative organization of the 16-bank computing platform 170 of FIG. 1(e), in which modularized logic circuits 172 are distributed among modularized memory circuits 171, according to one embodiment of the present invention.

FIG. 1(j) shows circuit 190 in which integrated circuits 120a and 120b implement data-intensive in-memory computations and massive, parallel searches in CAMs, respectively, according to one embodiment of the present invention.

FIG. 2(a) illustrates generally hybrid bonded memory chip 101 and companion chip 102 of integrated circuit 120, according to one embodiment of the present invention; in this embodiment, memory chip and companion chip 102 are bonded in a “flip-chip” orientation.

FIG. 2(b) illustrates in greater detail array structures 202-a and 202b, which are representative of any two adjacent array structures in array structures 202-(l,l) to 202-(n,m).

FIG. 2(c) illustrates generally hybrid bonded memory chip 101 and companion chip 102 of integrated circuit 120, according to another embodiment of the present invention; in this embodiment, memory chip and companion chip 102 are bonded in a “stacked” orientation.

FIG. 2(d) illustrates generally hybrid bonded memory chip 101 and companion chip 102 of integrated circuit 120,

according to a third embodiment of the present invention; in this third embodiment, memory chip 101 includes VNOR memory arrays.

FIG. 2(e) illustrates generally hybrid bonded memory chip 101 and companion chip 102 of integrated circuit 120, according to a fourth embodiment of the present invention; in this fourth embodiment, memory chip 101 includes VNOR memory arrays and vertical thin-film transistors (TFT).

FIG. 3 shows a portion of integrated circuit 120 of FIG. 2(a) in greater detail.

FIG. 4 shows a top view of companion chip 102, showing stripe 203 of hybrid bonds and metal layers 206.

FIG. 5(a) illustrates sense amplifiers, latches, SRAM arrays and various logic circuitry formed at surface 212 of substrate 211 in companion chip 102, according to a first embodiment of the present invention.

FIG. 5(b) shows sense amplifiers, latches, SRAM arrays and various logic circuitry at surface 212 of substrate 211 of companion chip 102, according to a second embodiment of the present invention.

FIG. 6 generally illustrates memory module 600, according to one embodiment of the present invention, with memory chipsets 601-0 to 601-15 each including memory chip 101 and its companion chip 102 bonded together; the memory module may be provided in the format of a dual-in-line memory module (DIMM).

FIG. 7 illustrates integrated circuit 160, which includes memory chipset 120—formed by memory chip 101 and companion chip 102—and non-memory chip 161.

FIG. 8 shows memory chipset 120 and non-memory chip 161 interconnected by silicon interposer 801.

FIG. 9 is a schematic representation of computing system 900, which may be a subsystem within a larger host system (e.g., host system 603 of FIG. 6).

FIG. 10 is a schematic representation of memory chipset 125, which is provided battery 1001 on companion chip 102.

FIG. 11 schematically illustrates a paging system using the fast memory circuits (e.g., SRAM circuits) and the quasi-volatile memory circuits of integrated circuit 120, in accordance with one embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

According to one embodiment of the present invention, an integrated circuit may be formed by combining high-density, quasi-volatile memory circuits, or non-volatile memory circuits, formed on a first semiconductor die (“memory chip”), and faster memory circuits (e.g., SRAM, DRAM, eDRAM, MRAM, eMRAM, PCM or any other suitable memory circuits) formed on a second semiconductor die (“companion die”). The quasi-volatile memory circuits or the non-volatile memory circuits on the memory chip are preferably built for high density, such as achieved through three-dimensional construction. In contrast, the faster memory circuits on the companion chip are preferably built for high performance, such as achieved through more advanced logic process nodes. The memory chip and the companion chip may be brought together by high-density hybrid bonding, for example.

Of importance, in one embodiment of the present invention, both the memory chip and the companion chip are organized in modular blocks, which are colloquially referred to as “tiles.” In that embodiment, the tiles of the memory chip and the tiles of the companion chip have a one-to-one to correspondence. Each tile area in the companion chip—

which is equivalent in area to a corresponding tile in the memory chip—provides the sense amplifiers and other logic support circuitry for the quasi-volatile memory circuits in the corresponding tile. In addition, each tile in companion chip includes fast memory circuits (e.g., SRAM circuits) placed within specific “pocket” areas on the tile. As a result, the corresponding tiles in the memory chip and the companion chip form a very high-density, very low-latency heterogeneous memory circuit (i.e., the three-dimensional construction of the memory circuits of the memory chip (e.g., quasi-volatile memory circuits) providing the high density, and the fast memory circuits providing very low-latency (e.g., SRAM circuits)). The memory circuits on the memory chip may include 3-D NAND, 3-D PCM, 3-D HNOR memory, 3-D VNOR memory or other suitable non-volatile or quasi volatile memory circuit types. The memory circuits on the companion chip may include volatile memory circuits (e.g. SRAM or DRAM), or high-performance, non-volatile memory circuits (e.g. MRAM, ST-MRAM or FRAM), or any suitable combination of these types of memory circuits.

According to one embodiment of the present invention, high-performance, low-voltage transistors are provided on the companion chip, rather than the memory chip, so as (i) to avoid degradation of the high-performance, low-voltage logic transistors during thermal cycles in the manufacturing of the memory arrays on the memory chip, and (ii) to benefit from advanced manufacturing nodes optimized for their production. As the low-voltage transistors form sense amplifiers, registers or data latches, high-performance data path circuits, input and output interfaces, error-correction circuits (ECCs), and fast logic circuits (e.g., the low-voltage decoders and multiplexers, state machines and sequencers, and input and output circuits) that can best take advantage of manufacturing process nodes that are one or more generations more advanced—albeit more costly—than the manufacturing process nodes that are capable of also manufacturing the high-voltage and medium-voltage transistors. In addition, depending on the intended application or the desired manufacturing technology, the memory chip may be hybrid bonded to a companion chip specifically configured for that intended applications, or may be manufactured using that manufacturing process (e.g., a sufficiently advanced or cost-effective CMOS manufacturing process node). High-performance, low-voltage transistors are particularly susceptible to degradation during the thermal cycles in the manufacturing of the memory arrays. De-coupling the low-voltage transistors from the high-voltage and medium-voltage transistors by fabricating them on different chips provides an advantageous solution.

In one embodiment, while the medium-voltage and the high-voltage transistors are manufactured as CuA in the memory chip using, for example, 65-nm to 28-nm minimum design rules, the high-performance, low-voltage transistors on the companion chip may be implemented with the much faster and much denser 28-nm to under 5-nm low voltage-only design rules. Under this scheme, the companion chip not only provides the conventional support circuitry for the memory arrays in the memory chip, the density achievable using the more advanced manufacturing nodes allows inclusion of other circuitry (e.g., SRAM circuits, arithmetic and logic circuits, reduced instruction set computers (RISCs), and other suitable logic circuits) that may be effective, for example, in in-memory computation or near-memory applications. In addition, by providing low-voltage circuits in the companion chip, the CuA on the memory chip need only provide high voltage and medium-voltage transistors,

thereby allowing the memory chip to benefit from both a reduced die-size and a simpler manufacturing process, thereby resulting in a higher yield.

In this embodiment, both the word line-related circuits and their connections reside in the memory chip, without requiring word line-related hybrid-bond connections to the companion chip. Without such word line-related hybrid bond connections, the number of hybrid bonds required by this embodiment of the present invention is necessarily significantly less than that required by Lu’s 3-D NAND memory device, discussed above, which requires hybrid bond connections for all word line signals and all bit line signals to be received into or generated from support circuits (e.g., signal decoders) in the companion chip. The interconnection layers in the companion chip route the signals to and from the circuitry in substrate of the companion chip. Routing both word line-related and bit line-related signals to the companion chip thus results in leaving few hybrid bonds and routing tracks in the companion chip available for other signals or other uses. This problem is avoided in the present invention.

One embodiment of the present invention may be illustrated by FIG. 1(a). FIG. 1(a) shows integrated circuit **120**—which includes memory chip **101** and companion chip **102**, bonded together (e.g., using hybrid bonding)—operating under control or supervision by host processor **103**. (Other suitable bonding techniques include, for example, micro-bump or direct interconnect bonding.) In the detailed description below, bonded integrated circuit **120** may be referred to as a “memory chipset.” Host processor **103** may be, for example, a conventional central processing unit (CPU), a graphics processing unit (GPU), a field programmable gate array (FPGA) or a memory controller. As shown in FIG. 1(a), memory chip **101** may include any quasi-volatile or non-volatile memory circuit, e.g., any of the types described in the Non-provisional Applications II and III. Examples of these quasi-volatile memory circuits include HNOR memory string arrays and VNOR memory string arrays. The quasi-volatile memory circuit may include numerous 3-dimensional arrays of thin-film storage transistors formed above a monocrystalline semiconductor substrate. The semiconductor substrate may also have formed therein suitable support circuitry (CuA), such as voltage sources for generating signals used in reading, programming or erase operations. As described below, low-voltage, fast logic circuits, sense amplifiers and other support circuitry for the quasi-volatile memory circuit may be implemented in companion chip **102**.

The high-density memory arrays on memory chip **101**, when implemented using quasi-volatile memory circuits, provide the benefit of high endurance. In read-intensive applications, however, the high-density memory arrays on memory chip **101** may be implemented by non-volatile memory circuits, or a combination of quasi-volatile memory circuits and non-volatile memory circuits. In that combination, non-volatile memory circuits are used to store data that is rarely changed and for which long-term retention is more important than high endurance. Examples of three-dimensional non-volatile and quasi-volatile memory circuits that can be used on memory chip **101** are described, for example, in Non-provisional Applications II and III.

Companion chip **102** may include fast memory circuits **107**, shown in FIG. 1(a) as modularized fast memory circuits **107-1**, **107-2**, . . . , **107-n**. Support circuitry for the quasi-volatile memory circuits and fast memory circuits **107** on companion chip **102** may be interconnected to the CuA on memory chip **101** using hybrid bonds. FIG. 1(a) shows

each of the modularized fast memory circuits tightly coupled to corresponding memory tiles of memory chip **101**. For example, in memory chip **101**, memory banks **110-1**, **110-2**, . . . , **110-*n*** (i.e., memory banks **bank[0]**, **bank[1]**, . . . , **bank[*n*]**), each of which may be a bank of quasi-volatile or non-volatile memory cells, are shown connected in close physical proximity, respectively, to modularized fast memory circuits **107-1**, **107-2**, . . . , **107-*n*** by, for example, hybrid bonds **111-1**, **111-2**, . . . , **111-*n***. In one embodiment, each modularized fast memory circuit on companion chip **102** is tightly coupled to a corresponding memory tile in memory chip **101**. Therefore, modularized fast memory circuits **107** become integral to the corresponding quasi-volatile or non-volatile memory banks **110**. In a practical implementation, memory chip **101** and companion chip **102** would be bonded to each other such that the least resistance result in the conductors (e.g., hybrid bonds **111**) between memory banks **110** in memory chip **101** and fast memory circuits **107** on companion chip **101**. As shown in FIG. 1(a), logic circuits **106** may also be modularized and laid out as modularized logic circuits **106-1**, **106-2**, . . . , **106-*n***, each being associated through close proximity and a corresponding one of low resistivity interconnect conductors **112-1**, **112-2**, . . . , **112-*n*** with a corresponding one of modularized fast memory circuits **107-1**, **107-2**, . . . , **107-*n***, which support the operations of their respective modularized logic circuits. Modularized logic circuits **106-1**, **106-2**, . . . , **106-*n*** may be any suitable logic circuits, such as multiplexers, adders, multipliers, Boolean logic operators, RISC processors, math-coprocessors, and FPGAs. Such modularized logic circuits **106** operating in conjunction with their associated modularized memory circuits **107** form what are sometimes referred to as “in-memory compute” elements. In-memory compute elements provide computational operations that are dominant in neural networks widely used in many machine learning, classification, and other artificial intelligence (AI) applications. In one embodiment, the computational complexity required of each of logic circuits **106** may be sufficient to call for implementing an embedded processor (e.g. a RISC processor, a math co-processor, or a micro-controller).

As shown in FIG. 1(a), other control circuitry and data paths, indicated generally as control and data circuits **108** may also be provided. Control and data circuits **108**, logic circuits **106**, volatile memory **107** and, through bonding pads of hybrid bonds **111**, circuitry on memory chip **101** are interconnected on companion chip **102** through various interconnect conductors **112**, **113** and **114**, and interconnection fabric **105**. Companion chip **102** communicates with host processor or controller **103** over input and output interface circuits **109**. Processor or controller **103** may be provided on a separate integrated circuit. Input and output interface **104** may be an industry-standard memory interface (e.g., DDR4, DDR5, or PCIe), through-silicon vias (TSVs), micro-bump or direct interconnects, or a second set of hybrid bonds.

In this embodiment, the 3-dimensional memory arrays and their associated CuA in memory chip **101** are organized in modular building blocks that are colloquially referred to as “tiles,” which are laid out over the semiconductor substrate in a 2-dimensional formation. Each tile may implement one or more 3-dimensional memory arrays, and the bit lines and the word lines used to access the tile’s memory arrays. As the word lines and the bit lines to access the tile’s 3-dimensional memory arrays are provided within the tile, their necessarily short lengths incur significantly less impedance than if they were routed over longer distances over the

semiconductor die. The lesser impedance facilitates lower read and write latencies to the memory cells in the memory array. In earlier tile implementations, the control circuitry, including drivers, decoders, multiplexers are provided in the CuA under the tile’s memory arrays. However, as mentioned above, a portion of the control circuitry (e.g., the sense amplifiers, registers and data latches) is provided in companion chip **102**, thereby significantly reducing the area required for the tile’s CuA. In this embodiment, the reduced area required to implement the CuA also result in a smaller tile.

In addition, the tiles may be organized into memory banks, with each bank having multiple rows of tiles and being addressable together by the same group of word lines. In one implementation, each row may have 18 tiles, each handling 2^{10} bits (“1 kbits”) of data input or output at a time, so as to handle a page of 2^{11} Bytes (“2-KByte”) of user data plus overhead (e.g., providing limited error correction and redundant spare tile capabilities). Some control structure (e.g., column or bit line decoders) may be shared among groups of multiple banks (“bank groups”). In one implementation, each bank group may be configured to have 2, 4, 8 or 16 banks.

In FIG. 1(a), modularized logic circuits **106-1**, **106-2**, . . . , **106-*n*** are each provided direct access through one of interconnect conductors **112-1**, **112-2**, . . . , **112-*n*** to a respective one of fast memory circuits **107-1**, **107-2**, . . . , **107-*n***. Depending on the computational need of a desired application, e.g., the computational power requirement on the modularized logic circuit **106**, or the nature of the data to be stored in fast memory circuits **107**, it may be more effective to have other organizations. For example, FIG. 1(b) shows an organization in which modularized logic circuits **106-1**, **106-2**, . . . , **106-(*n*-1)** are each provided direct access through two of interconnect conductors **112-1**, **112-2**, . . . , **112-*n*** to a respective two of fast memory circuits **107-1**, **107-2**, . . . , **107-*n***. Alternatively, in FIG. 1(c), a single modularized logic circuit **106** is provided direct access through interconnect conductors **112** and interconnection fabric **105** to each of fast memory circuits **107-1**, **107-2**, . . . , **107-*n***. Of course, the alternative configurations of FIGS. 1(a)-1(c) are not exhaustive, many variations and modifications are possible, based on the requirements of the desired application.

FIG. 1(d) shows a functional representation of one of fast memory circuits **107**, according to one embodiment of the present invention. FIG. 1(d) shows sense amplifiers **150**, which represents the sensed data values retrieved from a corresponding bank in quasi-volatile memory in memory chip **101** over hybrid bonds **110**. In each read activation cycle, each bank in memory chip **101** delivers to sense amplifiers **150** a fixed number of bits from each tile (e.g., 1024 bits). The data values are latched into master-slave register **151**, which allow the activated data to be held in the slave latch of master-slave register **151**, while the master latch of master-slave register **151** can be used for receiving data values from the next activation. Multiplexer **152**, in turn, selects a predetermined number of bits from the slave latch and place the selected bits onto compute data bus **154**—composed of true bus **154a** and complement bus **154b**, representing each bit in true and complement forms. Each bit and its complement on compute data bus **154** appears on the true and complement bit lines of a memory cell in fast memory array **153** (e.g., an SRAM array). Fast memory array **153** is mapped to the address space of the quasi-volatile memory, as seen from host processor **103**, for example. (As discussed below in conjunction with FIGS. 5a

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and **5b**, SRAM array **153** may reside in a set-aside portion of the address space, if desired). Word lines **155**, when enabled, write the data on compute data bus **154** into corresponding memory cells of fast memory array **153**.

Memory array **153** may be used as a bit-by-bit multiplier (without carry) which multiplies a first operand represented by the bits of word lines **155** and a second operand represented by the selected bits from the slave latch of master-slave register **151**. For example, in a matrix multiplication operation, the selected bits from the slave latch may represent elements in a row (or a portion of a row) in a matrix, and the bits on the word lines may represent a column (or a portion of a column) in the matrix. During an operation in multiplier mode, the enabled bits of word lines **155** writes the corresponding bits of the second operand into their corresponding memory cells, while the disabled bits in word lines **155** each trigger a reset signal that causes zero values to be written into the corresponding memory cells. The results stored in fast memory array **153** constitute the product terms of the multiplication operation. An adder and a carry circuit in a compute circuit **106** (e.g., one of arithmetic and logic circuits **106-1**, **106-2**, . . . , **106-4**) may provide a sum of the product terms to complete the multiplication operation. The result of the multiplication operation then may be written back from compute bus **154** back into fast memory array **153**. Multiplier mode is particularly advantageous in an application where matrix multiplications are heavily used, such as many AI applications.

FIG. **1(e)** shows a functional organization of memory bank groups BG(0)-BG(3) in companion chip **102**, according to one embodiment of present invention. As shown in FIG. **1(e)**, memory banks **180** of fast memory circuits **107** on companion chip **102** may be organized into portions **180a** and **180b**, both portions sharing data path control and input and output interface circuits **181**, representing control and data circuits **108** and input and output interface circuits **109** of each of FIGS. **1(a)-1(c)**. Portion **180a** includes bank groups BG(0) and BG(1), while portion **180b** includes bank groups BG(2) and BG(3), such that portions **180a** and **180b**, together, present four bank groups. In this embodiment, fast memory circuits **107** on companion chip **102** may service 64 Gbits of quasi-volatile memory cells on memory chip **101**. Each bank group is divided into two half-bank groups, indicated in FIG. **1(e)** as half-bank group **182-1a**, **182-2a**, **182-3a**, **182-4a**, **182-1b**, **182-2b**, **182-3b**, **182-4b**. Specifically, half-bank groups **182-1a** and **182-3a** form bank group BG[0], half-bank groups **182-2a** and **182-4a** form bank group BG[1], half-bank groups **182-1b** and **182-3b** form bank group BG[2], and half-bank groups **182-2b** and **182-4b** form bank group BG[3]. A general input/output bus, GIO bus **184** (indicated in FIG. **1(e)** by GIO buses **184a** and **184b**), allows access from input and output interface circuits **109**. In addition, for data transfer between bank groups, e.g., for computation using arithmetic and logic circuits **106**, 256-bit internal data bus DIO **183** (represented in FIG. **1(e)**, respectively, by 128-bit half-buses **183-1a** and **183-2a** in portion **180a** and 128-bit half-buses **183-1b** and **183-2b** in section **180b**) is provided. In this embodiment, each half-bank group may include four 8-tile wide half-banks, with each half-bank having 4-8 Mbits of fast memory cells. In this embodiment, GIO bus **183** delivers one page of data (2 Kbytes) over input and output interface **109** to host processor **103** under each cycle of an industry standard bus protocol (e.g., DDR5).

FIG. **1(f)** illustrates four half-banks **188a-188d** of a half-bank group (e.g., half-bank group **182-1a** of FIG. **1(e)**) on companion chip **102**, relative corresponding half-banks

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189a-189d of quasi-volatile memory circuits in memory chip **101**, according to one embodiment of the present invention. As shown in FIG. **1(f)**, each of half-banks **188a-188d** is bordered by a sense amplifier section (e.g., sense amplifier section **190-1**), on one side, and by an arithmetic and logic circuit section (e.g., arithmetic and logic section **191-1**) on the other side. Each sense amplifier services data retrieved over the hybrid bonds or micro-bumps from a corresponding half-bank of quasi-volatile memory cells (e.g., half-bank **189a**) in memory chip **101**. In one embodiment, sense amplifiers for 4096 bits of user data are provided in each half-bank. GIO bus **184**, in addition to allowing host access from input and output interface circuit **109**, also allows reading and writing between each half-bank of fast memory circuits of companion chip **102** and its corresponding half-bank of quasi-volatile memory circuits on memory chip **101**. In this manner, the fast memory circuits may serve as a cache to the corresponding quasi-volatile memory circuit or, independently, be used for storing frequently accessed data ("hot data"; e.g., data that is more than ten times more frequent than data stored in the quasi-volatile memory circuits), or as a storage of configuration or control data ("metadata") for the corresponding quasi-volatile memory circuits. Such metadata improves performance and reliability of the quasi-volatile memory circuits.

FIG. **1(g)** illustrates four half-banks **187a-187d** of a half-bank group (e.g., half-bank group **182-1a** of FIG. **1(e)**) on companion chip **102**, relative corresponding half-banks **189a-189d** of quasi-volatile memory circuits in memory chip **101**, according to an alternative embodiment of the present invention. As shown in FIG. **1(g)**, unlike the embodiment of FIG. **1(f)**, the half-bank **187a-187d** do not have identical configurations. The sense amplifiers in each of half-bank **187a-187d** are provided as sense amplifier sections on both sides of each half-bank (e.g., sense amplifier sections **190-1a** and **190-1b** in half-bank **187a**). Rather than providing arithmetic and logic circuits in each half-bank, arithmetic and logic circuits are concentrated in half-bank **187-d**. Other than its configuration, this alternative embodiment operates in the same manner as that described above in conjunction with the embodiment of FIG. **1(f)**. For some applications, this alternative embodiment may provide comparable or better performance than the embodiment of FIG. **1(f)**. For other applications, the embodiment of FIG. **1(f)** may provide better performance than the embodiment of FIG. **1(g)**.

As shown in FIGS. **1(e)** and **1(f)**, each half bank within each half-bank group (e.g., half-bank **188a** of half-bank group **182-1a**) is provided access to a compute bus (indicated generally by compute bus **184**), which is a bus shared between sense amplifier section, the fast memory circuits, and the arithmetic and logic circuits. In one embodiment, the compute data bus is 256-bits wide per tile column, with each half-bank group being eight tiles wide. (Of course, the widths of the compute data bus and the half-bank group may vary, depending on the requirements of the intended application.) Accordingly, a significant on-chip data bandwidth is provided within the half-bank group for data transfer between the sense amplifier section (which delivers the data read from the quasi-volatile memory circuits in the memory chip), the fast memory circuits and the arithmetic and logic circuits. In this manner, large amount of data may be streamed into the fast memory circuits as operands for arithmetic and logic operations with other operands that are other data or previous computational results that have already been stored in the fast memory circuits or in the quasi-volatile memory circuits. For example, in an AI appli-

cation, data may be stored in quasi-volatile memory and output through the sense amplification section during a read operation. The data then can be used, together with the weights stored in the fast memory circuits, to perform matrix multiplication, for example, using the on-chip arithmetic and logic circuits and the compute bus. This is in stark contrast with the practice in the prior art, which requires transferring data into or out of DRAM to processors (e.g., CPUs or GPUs). Under the embodiments of the present invention, such computations may be carried out without data transfer into or out of memory or the companion chip to the CPU or GPU.

Compute bus **184** enables massively parallel computational operations (“in-memory computations”) to be performed, without operand fetching and resulting storing operations involving a host interface bus. In this embodiment, as each bank group includes four banks, four sets of in-memory computations may be carried out in parallel in each bank group. Each tile column may be configured for the same or different in-memory computation from the other tile columns. The results of these in-memory computations may then be sent to the host over the input and output interface. The in-memory computations carried out simultaneously may be independent or may be parts of a coordinated computation (i.e., an in-memory computation for each bank may involve an entire page of data). These in-memory computations not only significantly improve power and performance, they make integrated circuit **120** particularly advantageous to many applications, such as many AI applications previously deemed intractable. For example, neural networks may be implemented using in-memory computations, using input data fetched from the quasi-volatile memory circuits together with the weights of the neurons and the intermediate results that are already stored or available in time from the fast memory circuits. As another example, recursive computations (e.g., those involved in recursive neural networks) may also be implemented by in-memory computations. With a quasi-volatile memory (e.g., 64 Gbits) on memory **101** and a large amount of on-chip fast memory circuits (e.g., 64 Gbits of SRAM) on companion chip **102**, their combination (i.e., integrated circuit **120**) enables both heretofore unachievable performance for existing applications and heretofore intractable computational applications.

Companion chip **102** makes integrated circuit **120** essentially a computing platform with high density (e.g., greater than 64 GBytes) quasi-volatile or non-volatile memory available at a much greater bandwidth relative to conventional high-performance computing platforms that use DRAM modules (e.g., HBM modules) connected to a host processor over interposer connections. FIG. **1(h)** illustrates functional configuration of a 16-bank computing platform **170**, including computing banks **170-1**, **170-2**, . . . , **170-16**, based on organization such as that described above in conjunction with FIG. **1(e)**, according to one embodiment of the present invention. As shown in FIG. **1(h)**, representative computing bank **170-1** includes representative modularized memory circuits **171-1**, **171-2** and **171-3** (e.g., SRAM circuits) that constitute a memory bank, such as any of the memory banks in FIG. **1(e)**, discussed above. In addition, computing bank **170-1** also includes representative modularized logic circuits **172**, connected to modularized memory circuits **171-1**, **171-2** and **171-3** over local compute bus **173** (e.g., compute bus **154**, described above). (The number of modularized memory circuits in each bank in FIG. **1(h)** is provided merely for illustrative purpose; any suitable number of modularized memory or logic circuits are possible.)

Local bus **173** in each of computing banks **170-1** to **170-16** has access to an intra-bank data bus (e.g., GIO bus **184** or DIG bus **183**, described above) to allow data transfer between computing banks. In this configuration, modularized logic circuits **172** may form any suitable computing circuit, such as an ALU core, a GPU core, or any suitable embedded controller or microprocessor. Modularized logic circuits **172** may be implemented, for example, by FPGAs. In the configuration of FIG. **1(h)**, computing bank **170-1** may form a CPU with a 16-Mbyte SRAM cache that supports a 16-GB memory provided by the quasi-volatile or non-volatile memory of memory chip **101**. One advantage of computing banks **170** arises from having modularized logic circuits **172** (e.g., the ALU or GPU core) in close proximity with fast memory circuits **171-1**, **171-2** and **171-3**, facilitated by local compute bus **173**. In fact, an even greater advantage may be achieved by distributing the modularized logic circuits among the modularized fast memory circuits, such as illustrated by FIG. **1(i)**, so as to provide greater proximity between the modularized memory circuits and modularized the logic circuits. Data transfers between computing banks may be carried out in interbank data bus **175**.

As shown in FIG. **1(i)**, each of computing banks **170-1**, . . . , **170-16** includes modularized fast memory circuits **171-1**, **171-2**, . . . , **171-n** and modularized logic circuits **172-1**, **172-2**, . . . , **172-n**. In addition to intra-bank local compute bus **173**, modularized data buses **174-1**, **174-2**, . . . , **174-n** may be provided, each allowing data transfer between a modularized memory circuit and a modularized logic circuit adjacent to it. Thus, each modularized logic circuit may connect to a processor core in proximity.

The 16-bank computing platform may be configured to operate in a pipelined manner. For example, a deep neural network may include many layers. In one embodiment, one may use one computing bank for each layer of such a deep neural network. The weight matrix for the neurons in that layer of neural network may be stored in the fast memory circuits of the computing bank. When computation of a layer of the neural network is complete, its results are forwarded over to the next computing bank. The forwarding of data from one computing bank to another may be carried out in a synchronous manner, i.e., at a specified edge of a clock signal. This way, after an initial latency of 16 cycles, results for deep neural network may emerge every cycle thereafter. For this kind of computation, a conventional processor is limited by the total amount of data that can be placed in the fast memory circuits (e.g., SRAM) and then must go off-chip to fetch new data from DRAMs.

Non-provisional Application IV discloses logical functions that can be implemented using NOR memory strings, such as a content-addressable memory (CAM). A CAM allows parallel search of data. Because of the high-density achievable in memory chip **101**, a CAM may be implemented on integrated circuit **120** to enable massive, parallel search data, as disclosed in Non-provisional Application IV. FIG. **1(j)** shows circuit **190** in which integrated circuits **120a** and **120b**—both copies of integrated circuit **120**, described above—implement data-intensive in-memory computations and massive, parallel searches in CAMs, respectively. Integrated circuits **120a** and **120b** are both controlled by host processor **103** over memory interface **104**. For example, integrated circuit **120a** may be tasked with highly data-intensive computations, such as image classification. The results of the data-intensive computations may be transferred, under control of host processor **103** over memory interface bus **104** to integrated **120b**, where a massively parallel search may be carried out of an image database

stored in CAM circuits in memory chip 101. For the reasons already stated above and in Non-provisional Application IV, both these operations, individually and in combination, are expected to deliver very fast execution. One also can envision using many copies of integrated circuits 120, with some programmed for logic functions and the rest implementing CAMs. In that configuration, the logic function integrated circuits may be programmed to perform various computation tasks in parallel, or in one or more pipelines, with their results provided over one or more high-bandwidth memory interface buses for parallel searches.

FIG. 2(a) illustrates generally “flip-chip” or “face-to-face” bonded memory chip 101 and companion chip 102 of integrated circuit 120. In the embodiment of FIG. 2(a), companion chip 102—rather than memory chip 101—implements sense amplifiers (represented in FIG. 2(a) as some of circuit elements 208-1 to 208-n) that support the operations of the quasi-volatile or non-volatile memory arrays 202 of memory chip 101. Companion chip 101 also implements bit line control logic circuits at or near surface 212 of substrate 211 (represented in FIG. 2(a) by some of circuit elements 208-1, . . . , 208-n). Companion chip 102 also may route external high voltage signals (not shown) from the CuA of memory chip 101, supplying arrays 202 of quasi-volatile or non-volatile storage cells in memory chip 101. For example, high-voltage bit line-select (BLSEL) transistors are provided in the CuA of memory chip 101, each multiplexing multiple bit line signals of quasi-volatile memory array 202 onto a bit line-internal (BLI) node, which is then routed over a hybrid bond as an input signal to a corresponding sense amplifier on companion chip 102. In companion chip 102, the BLI node is connected by a conductor-filled via (represented by via 215 in FIG. 2(a)) to an input terminal of a sense amplifier, represented in FIG. 2(a) respectively by via 217 and circuit element 208-2.

The sense amplifiers and their associated data latches, which are formed by high-performance, low-voltage transistors on companion chip 102 using an advanced manufacturing process node that is optimized to CMOS logic technology, and are not exposed to the thermal cycles in the formation of the quasi-volatile memory arrays of memory chip 101, would suffer no performance degradation due to the thermal cycles. As the additional capacitance of the BLI node is very small (e.g., less than 2%), such a capacitance has no substantial impact on either the sense amplifier performance or operation. Under this arrangement, the CuA on memory chip 101 implements high-voltage word line and bit line decoders, drivers and multiplexers. As a result, the “division of labor” between memory chip 101 and companion chip 102 not only reduces the area requirement on the CuA of memory chip 101, the multiplexing of signals through the BLI nodes greatly reduces the number of hybrid bonds required to route bit line signals to companion chip 102. This is in stark contrast to, for example, the use of hybrid bonds for routing bit line signals, as taught by Lu, discussed above. In this embodiment, rather than ~20,000 hybrid bonds per tile required without multiplexing (as taught in Lu), about ~1K hybrid bonds are required in each tile to route the bit line signals to companion chip 102, while enjoying the advantage of high signal integrity that results from not exposing the high-performance, low voltage circuits (e.g., the sense amplifiers) in the thermal cycles in the manufacturing process of the quasi-volatile memory arrays. The significant reduction in the number of hybrid bonds needed to route signals to companion chip 102 substantially releases a significant number of routing channels in the metal interconnect layers of companion chip 102. Not

implementing the high-performance, low-voltage logic circuits in memory chip 101 also reduces the number of masking steps required in the fabrication of memory chip 101, resulting in a simpler manufacturing process (i.e., higher yield) and lower wafer processing cost in producing memory chip 101.

Having sense amplifiers for memory array 202 of memory chip 101 and high-performance, low-voltage fast memory circuits 107 and logic circuits 106 all in close proximity with each other on the companion chip 102 provides the advantages of: (i) allowing these circuits to be manufactured under a process optimized for their performance, (ii) avoiding power-hungry and time-consuming computational operations that bring data from memory chip 101 to companion chip 102 and back to memory chip 101 again, (iii) providing greater noise immunity from high-voltage circuitry, which still resides on memory chip 101, thereby resulting in greater sensing sensitivity; (iv) leveraging the fast memory circuits and the sense amplifiers in the companion chip to carry out write operations (i.e., both programming and erase) in parallel in the quasi-volatile memory circuits (i.e., servicing read operations from the fast memory circuits, while a write operation involving data on the same page is carried out in parallel in the quasi-volatile memory circuits); and (v) leveraging the fast memory circuits and the sense amplifiers to monitor the health of quasi-volatile memory circuits, so as to improve reliability and endurance of the quasi-volatile memory circuits.

In one embodiment, memory chip 101 has a 64-Gbit storage capacity in the three-dimensional quasi-volatile memory arrays, segmented into 1,024 tiles, each tile having 64 Mbit of random access quasi-volatile memory cells, with its supporting circuits in the CuA (except for the sense amplifiers). Read latency to a location in the quasi-volatile memory array is approximately 100 nanoseconds, with an endurance of approximately 10^{10} programming and erase cycles. In that embodiment, each tile in memory chip 101 is separately connected by hybrid-bonded to a corresponding one of 1024 SRAM modules on companion chip 102. On companion chip 102, each tile has (i) 64 Kbits of SRAM cells and (ii) the sense amplifiers for supporting the quasi-volatile memory cells in the corresponding tile of memory chip 101. Read latency to a location in the SRAM cells of the tile is approximately 25 nanoseconds, with an essentially unlimited endurance. Having the SRAM modules on companion chip 102 serve as a fast cache memory, uniquely mapped to quasi-volatile memory arrays in corresponding designated tiles, results in a heterogeneous memory circuit that can deliver the best advantages of both memory types, i.e., (i) the significantly higher density of the quasi-volatile memory cells and (ii) the significantly faster read access times and the significantly higher endurance in the SRAM circuits. Thus, where relying solely on SRAM circuits may be too costly for applications operating on large data sets, or where relying solely on quasi-volatile memory circuits may be too slow or have an endurance that is inadequate to support high-frequency, read-intensive or write-intensive applications, the heterogeneous memory circuit that combines the memory types can provide a superior solution. The present invention includes circuitry and methods for allocating data between the fast memory circuits (e.g. SRAM) and the slower memory circuits (e.g. quasi-volatile memory) and moving data between one type of memory circuits and the other type of memory circuits without host involvement.

As shown in FIG. 2(a), memory chip 101 includes an n by m formation of tiles, each tile having a CuA structure and an associated array structure. Thus, FIG. 2(a) shows CuA

structures **201-(l,l)** to **201-(n,m)** and array structures **202-(l,l)** to **202-(n,m)**. Each CuA structure may include, for example, various voltage sources and various high-voltage and medium-voltage analog and logic circuits to support its corresponding tile. On the side of this formation of tiles are provided sequence and control modules **209-1** to **209-n**, each including sequencers (Seq) and bit line and word line control circuits for memory banks (BCU). As discussed above, each array structure consists of a 3-dimensional array of storage cells, organized as quasi-volatile or non-volatile NOR memory strings, and a staircase structure, which allows electrical access to the common drain region or bit line of each NOR memory string. FIG. 2(b) illustrates in greater detail array structures **202-a** and **202-b**, which are representative of any two adjacent array structures in array structures **202-(l,l)** to **202-(n,m)**. As shown in FIG. 2(b), array structure **202-a** and **202-b** each include an array of storage cells (exemplified by arrays **251a** and **251b**, respectively) and, on its opposite sides, staircases (exemplified by staircases **252a** and **252b**). FIG. 2(b) also shows signals from the CuA of memory chip **101** being routed through conductor-filled vias **254a** and **254b** to hybrid bonds **253a** and **253b** over interconnect conductor layer **256**, with sections **256a** and **256b** overlapping, staircases **2512** and **252b**, respectively.

Memory chip **101** and companion chip **102** are bonded by stripes **203-1** to **203-n** of hybrid bonds, each stripe running along the word-line (WL) direction, with each stripe of hybrid bonds provided above the space between the storage cell arrays of adjacent array structures, overlapping their respective staircases. These hybrid bonds connect signals traveling “vertically” (i.e., substantially perpendicular to the surfaces of the semiconductor substrates) through conductor-filled vias. In one embodiment, where desirable, signals connected by hybrid bonds between the memory chip and the companion chip are multiplexed and demultiplexed to share and increase the effective number of interconnections by hybrid bonds and to overcome the density limitations of current hybrid bond technology. FIG. 2(a) also shows metal layers **204-207** in companion chip **102**. Metal layer **204** provides an interconnection layer that is used to distribute signals to destinations in both memory chip **101** and companion chip **102**, including high voltage signals originating from CuA in memory chip **101**. Metal layer **205** provides a substantial ground plane that shields other circuits in companion chip **102** from interference by these high voltage signals. Metal layer **206** provides parallel interconnect conductors (“feed-thru conductors”) each extending along the bit-line (BL) direction to allow bit lines signals to be routed to a second interconnection network **207**, which has interconnection conductors running along the WL direction.

More specifically, hybrid bonds **203-1** to **203-n** connect bit lines from array structures **202-(l,l)** to **202-(n,m)** in memory chip **101** to sense amplifiers at surface **212** of substrate **211** in companion chip **102** and between the circuitries in the CuA of memory chip **101** and the circuitry at surface of substrate **211** in companion chip **102**. Hybrid bonds **203-1** to **203-n** also routes the high voltage signals from the voltage sources at surface **212** of the semiconductor substrate in memory chip **101** to other portions of memory chip **101** through metal layer **204** in companion chip **102**. Substrate **211** may be a semiconductor wafer that is thinned after formation of the circuitry of companion chip **102** to an insulator layer, e.g., silicon oxide layer. Alternatively, substrate **211** may be formed by implanting oxygen atoms into the semiconductor wafer to form an oxide layer, after annealing. After formation of the circuitry of companion

chip **102** at surface **212**, substrate **211** may be separated from the semiconductor wafer mechanically. Substrate **211** is referred to as a silicon-on-insulator (SOI) substrate. Bonding pads **210-1** to **210-n** may then be formed on the cleaved surface **213**.

FIG. 2(a) also shows bonding pads **210-1** to **210-n** on surface **213** of substrate **211**, opposite to surface **212**, where circuit elements **208-1** to **208-n** are formed. Bonding pads **210-1** and **210-n** are each provided to allow access to signals from the circuitry formed at surface **212** of substrate **211** through TSVs, such as those shown in FIG. 2(a) as TSVs **214-1** to **214-n**. Bonding pads **210-1** to **210-n** may allow wafer-level or chip-level bonding to another substrate. Suitable bonding techniques may be hybrid bonding, direct interconnect or micro-bump bonding. In FIG. 2(a), for illustrative purpose, bonding pad **210-n** is represented by a bonding pad suitable for hybrid bonding. Bonding pad **210-1** is represented by a micro-bump suitable for micro-bump bonding.

FIG. 2(c) illustrates generally hybrid bonded memory chip **101** and companion chip **102** of integrated circuit **120**, according to another embodiment of the present invention; in this embodiment, memory chip **101** and companion chip **102** are bonded in a “stacked” orientation. As shown in FIG. 2(c), memory chip **101** and companion chip **102** each contain substantially the same circuitry as described above in conjunction with FIG. 2(a), except that bonding pads for hybrid bonding (or micro-bumps for micro-bump bonding, as the case may be) are formed on the “backside” of substrate **211**. This is achieved, for example, by having companion chip **102** fabricated on an SOI substrate, which is thinned down sufficiently (e.g., down to 3 microns or thinner). Connectors (e.g., bonding pads or micro-bumps) are then formed on surface **213** of substrate **211** to mate by hybrid bonding (or micro-bump bonding) with corresponding connectors on memory chip **101**. Connectors on surface **213** of substrate **211** are connected to circuitry at surface **212** by miniaturized high-density TSVs through conductor-filled vias through substrate **211**. Relative to the “flip-chip” embodiment shown in FIG. 2(a), this embodiment has the advantage that the complexity of signal routings in metal layer **204**, **205**, **206**, and **207** (e.g., “feed-thru” routing in metal layers **206** and **207**) may be significantly simplified, or substantially avoided.

In FIGS. 2(a) and 2(c), memory chip **101** implements HNOR memory string arrays. The present invention also may be practiced with memory **101** implementing quasi-volatile or non-volatile VNOR memory strings arrays. Various embodiments of VNOR memory string arrays are described, for example, in Non-provisional Application III. FIG. 2(d) illustrates generally hybrid bonded memory chip **101** and companion chip **102** of integrated circuit **120**, according to a third embodiment of the present invention; in this third embodiment, memory chip **101** includes VNOR memory string arrays. As shown in FIG. 2(d) row **220** in a tile of one or more quasi-volatile or non-volatile VNOR memory string arrays includes memory string-pairs **228-1**, **228-2**, . . . , and **228-n**, with two VNOR memory strings formed on opposite sides of each memory string-pair.

As shown in FIG. 2(d), the VNOR memory strings in each memory string-pair share a common source line and a common bit line, indicated in FIG. 2(d) by bit lines (BLs) **222-1**, **222-2**, . . . , and **222-n** and source lines (SLs) **223-1**, **223-2**, . . . , and **223-n**, respectively. On both sides of each of memory string-pairs **228-1**, **228-2**, . . . , and **228-n** between the common bit line and the common source line are formed two channel regions each isolated from a stack

of word line conductors by a charge-trapping layer. In FIG. 2(d), one stack of word line conductors is represented by word line conductors **221-1**, **221.20**, . . . , and **221-m**. Across row **220**, the common source lines and the common bit lines of the memory string-pairs alternate between the front and the back portions of row **220**. A pair of conductors (“global bit lines”) **224-1** and **224-2** connect the common bit lines of row **220** at the front and the back of memory string-pairs **228-1**, **228-2**, . . . , and **228-n**. In this embodiment, the common source lines **223-1**, **223-2**, . . . , **223-n** are each pre-charged by voltage applied to the associated one of common bit lines **222-1**, . . . , **222-n**, or by hardware connections (not shown) to voltage sources in the CuA of memory chip **101**, as described in Non-provisional Application III.

Bit line selector circuits **225** each connected to global bit lines of multiple rows of VNOR memory strings in the tile are provided in the CuA underneath VNOR memory string array to select a signal from one of the global bit lines **224-1** and **224-2** in the tile. Bit line selection circuits **225** perform substantially the same function as the multiplexers that select from bit line signals to provide selected bit line signal BLI described above in conjunction with FIG. 2(a). In this embodiment of FIG. 2(d), the selected signal is provided to bit line signal BLI represented by conductor-filled via **226**, which is connected to one of bonding pads (or micro-bumps) **227** at the bonding surface of memory chip **101**. Bonding pads (or micro-bumps) **227** connect with corresponding bonding pads (or micro-bumps) in companion chip **102** by hybrid bonding (or micro-bump bonding) in substantially the same manner as described above in conjunction with FIG. 2(a).

FIG. 2(e) illustrates generally hybrid bonded memory chip **101** and companion chip **102** of integrated circuit **120**, according to a fourth embodiment of the present invention; in this fourth embodiment, memory chip **101** includes VNOR memory string arrays and vertical thin-film transistor (TFT) that serve as an additional layer of bit line selection circuits. In FIG. 2(e), an additional conductor layer of global bit lines, represented by global bit lines **224-1** and **224-2** are provided in a metal layer (“global bit line layer **224**”) above the VNOR memory string array. In this embodiment, these additional global bit lines are not connected to bonding pads **227** by the bit line selector circuits in the CuA of memory chip, but by vertical TFTs, represented in FIG. 2(e) by vertical TFTs **229** formed above global bit line layer **224**. Vertical TFTs being used for bit line selection are described in the Provisional application. Having source line selection circuits **230** and bit line selection circuits in vertical TFTs **229** allow greater flexibility in routing bit line signals through the BLI nodes to sense amplifiers in companion chip **102**. As the number of hybrid bonds required for this routing may be reduced, the footprints for memory chip **101** and companion chip **102** may be reduced, thereby resulting in the advantages of a denser circuit. The vertical TFTs also may be used in HNOR memory string arrays to efficiently select and route bit lines to companion chip **102**.

FIG. 3 shows a portion of integrated circuit **120** of FIG. 2(a) in greater detail. As shown in FIG. 3, stripes **203-1**, **203-2** and **203-3** of hybrid bonds are provided adjacent to array structures **202-(1,1)** and **202-(2,1)**, which are representative of any two adjacent array structures **202-(l,l)** to **202-(n,m)** of FIG. 2(a). Some of the signals connected by stripes **203-1**, **203-2** and **203-3** of hybrid bonds are routed by conductor-filled vias to the circuitry at surface **212** of substrate **211** vertically through openings in metal layers **204-207** of companion chip **102**. Other signals are fanned

out by feed-thru metal layer **206**. As discussed above, metal layer **204** also allows routing of high voltage signals back to memory chip **101**, as illustrated by signal path **302** that connects a signal in a conductor in metal layer **204** to array structure **202-(1,1)**. FIG. 3 also shows areas **301-1**, **301-2** and **301-3**, which are projections of stripes **203-1**, **203-2** and **203-3** onto the semiconductor substrate of companion chip **102**. The gaps (“pocket areas”) between adjacent pairs of areas **301-1**, **301-2** and **301-3** are relatively large areas on the semiconductor substrate of companion chip **102**.

FIG. 4 shows a top view of companion chip **102**, showing stripe **203** of hybrid bonds and metal layer **206**. As shown in FIG. 4, stripe **203** includes hybrid bonds **503**. Certain ones of hybrid bonds **503** are used for routing the BLI nodes which are connected in companion chip **102** by conductor-filled vias (“BLI vias”) **215**. Signals routed on metal layers **204-207** must route around (i.e., “feed-thru” routing) around BLI vias **215**, such as illustrated by conductor **505** on metal layer **206**, which is seen to “jog” around two of BLI-node vias **215**. Not shown in FIG. 4 are signal lines in metal layer **204** are provided to route the high voltage signals. Each high voltage signal is routed by a conductor between two grounded conductors on the same metal layer (i.e., metal layer **204**) which provide additional shielding (in addition to the ground plane in metal layer **205**, also not shown in FIG. 4). Interconnect conductors **501** are interconnect conductor in feed-thru metal layer **206**.

According to one embodiment of the present invention, the pocket areas may be used for circuitry that enable integrated circuit **120** capabilities not previously available to memory circuits. For example, FIG. 5(a) shows circuitry at surface **212** of substrate **211** of companion chip **102**, according to one embodiment of the present invention. FIG. 5(a) shows representative circuit module group **510a** and **510b** in circuitry at surface **212** of substrate **211** of companion chip **102** separated by an area (“pad area”) that provides input and output interfaces of integrated circuit **120** (e.g., data input and output buses for communication with host processor **103**). Each of circuit module group **510a** and **510b** includes a 2-dimensional array of circuit modules, with each column of circuit modules (i.e., along the WL direction) occupying the pocket areas between adjacent stripes of hybrid bonds. In FIG. 5a, each of circuit module groups **510a** and **510b** includes types **521** and **522** of circuit modules. Circuit module type **521** may be circuit modules each including volatile memory circuitry (e.g., SRAM arrays). Circuit module type **522** includes column decoder circuits servicing both adjacent memory bank groups in the volatile memory circuitry of the same column and quasi-volatile storage cells in corresponding array structures in memory chip **101** (i.e., specific tiles related by locality).

FIG. 5(a) also shows variation **531** and **532** of type **521** circuit modules. Each of variations **531** and **532** include one or more SRAM arrays **541** and sense amplifier and data latch circuitry **543**. The sense amplifiers and data latches may each be shared among multiple memory cells in the memory array using multiplexers. Variation **531** may implement a single-ported SRAM array, while variation **532** may implement a dual-ported SRAM array.

In one embodiment, all the SRAM arrays **541** in companion chip **102** may occupy a different address space than the quasi-volatile storage cells in memory chip **101**, as illustrated in address space map **550**. In address space map **550**, SRAM arrays **541** are mapped to lower addresses, while quasi-volatile storage cells in memory chip **101** are mapped to the higher addresses. Thus, the quasi-volatile storage cells and SRAM **541** together form an extended

address space, integrating and sharing data lines within the same memory bank. The extended address space enables read and write operations to be serviced from SRAM 541, while a programming, erase or a refresh operation is in progress in the quasi-volatile memory circuits.

Optionally, the circuit modules may also additionally implement arithmetic and logic circuitry 544 (e.g., adders, multipliers, dividers, subtractors, RISC processors, math co-processors, and logic gates, such as XOR). A circuit module with both SRAM array and arithmetic and logic circuitry are particularly suitable for implementing in-memory and near-memory computation desired in many applications, such as machine learning, classification, neural networks and other AI application. Because of much higher bandwidth between SRAM array 541 and arithmetic and logic circuitry 544—i.e., data retrieved from and written back to memory are routed between the memory and the processing units over on-chip signal routing, without the limited bandwidth of a conventional memory interface bus (the “von Neuman bottle neck”)—substantially greater performance is achieved, as compared with those of conventional processor architecture. With battery or capacity back-up power, the SRAM arrays retain its data even during a period of power loss, thereby allowing unlimited access to the same data without conflict with the need to perform refresh operations, which is particularly suitable for storing system data, as well as application and operating system software. In addition, recursive computation operations for training in AI applications may be performed using large storage capacity of the quasi-volatile memory circuits and fast SRAM circuits. Furthermore, the quasi-volatile memory circuits may be part of a larger memory with both quasi-volatile and non-volatile memory sections, with the non-volatile memory section storing weights that do not change frequently.

Alternatively, SRAM arrays 541 may each be used as a cache for quasi-volatile storage cells in corresponding array structures in corresponding memory banks. Because memory chip 101 and companion chip 102 are interconnected by hybrid bonds, which can be organized to provide high-bandwidth internal data buses (e.g., a 256-bit or 1024-bit wide bus per tile) between corresponding quasi-memory circuits of memory chip 101 and SRAM arrays in companion chip 102, To implement the cache function, circuitry may be provided in each circuit module to directly transfer data from the memory banks over these high-bandwidth internal data buses to the corresponding SRAM arrays (e.g., a page at a time). In one embodiment, each SRAM array has a storage capacity of 64 kbits and serves as a cache for a quasi-volatile memory circuit 64 Mbits. In that embodiment, a row of 16 tiles (plus overhead) are activated together to provide a 2-Kbyte page that is loaded or written together. In this manner, a single activation at the corresponding quasi-volatile memory bank prefetches a data page (after sensing at the sense amplifiers) into SRAM array 541. If host processor 103 accesses data at conventional cache-line sizes (e.g., 64 bytes) and with locality of reference, each prefetch can service many read accesses. If SRAM array 541 maintains multiple pages of a corresponding quasi-volatile memory bank in memory chip 101, the effective read latency of integrated circuit 120—amortizing the activation time of the quasi-volatile memory bank over many host accesses—approaches the read latency of the SRAM array. The activation time of an SRAM bank (e.g., 2 ns or less) is very short relative to the activation time of the corresponding quasi-volatile memory circuit. Furthermore, write operations may

be deferred until a page of the quasi-volatile memory bank cached in SRAM array 541 needs to be swapped out or “evicted”.

As it is preferred and sometimes required in quasi-volatile memory arrays to write or erase a page at a time, such deferred write of cached data from SRAM array 541 is particularly favored from both the performance and endurance points of view. From the performance point of view, amortizing the write access time of the quasi-volatile memory bank over many host computer accesses provides integrated circuit 120 SRAM circuit-like performance. As a result, with a multi-page cache in SRAM array 541, the performance of the combined volatile and quasi-volatile memory is effectively the performance of SRAM memory circuit. In addition, as SRAM arrays dissipate minimal power when not actively read or written, integrated circuit 120 with both SRAM and quasi-volatile memory circuits is very energy efficient. As data is mostly operated on and accessed in the SRAM circuits, this combination of SRAM and quasi-volatile memory circuits reduces power consumption because there are fewer read, write and erase operations performed on the quasi-volatile memory circuits. With fewer read, write and erase operations performed on the quasi-volatile memory circuits, the frequencies of erase-inhibit disturbs, write-inhibit disturbs, and read-disturbs in the quasi-volatile memory are correspondingly reduced. As well, greater endurance is achieved, as the quasi-volatile memory cells have significantly less exposure to the high-voltage electric field stress under write and erase operations.

As mentioned in Non-provisional Applications I and II, quasi-volatile memory circuits require refresh operations to retain data beyond their retention times (e.g., minutes). Naturally, when a data read operation is being performed on a page of memory cells at a time the page is due for a refresh operation, a “refresh conflict” arise. One of ordinary skill in the art would understand that a refresh conflict (e.g., those occurring in DRAMs) is sometimes resolved by stalling the read operation until the refresh operation is complete. Refresh conflicts are therefore an overhead cost that adversely effect memory performance. However, using the SRAM arrays as cache for corresponding quasi-volatile memory arrays in the memory circuit, read operations are likely serviced out of the SRAM cache, rather than requiring an access to the quasi-volatile memory circuits, thereby substantially avoiding most refresh conflicts. As the retention times of quasi-volatile memory circuits are already relatively longer than DRAMs, using an SRAM cache in conjunction with a quasi-volatile memory, as provided by the present invention, the effective performance that can be achieved likely surpasses that of conventional memory systems, such as DRAMs.

A cache in the prior art consists primarily of fast dedicated memory circuits (e.g., SRAM or SRAM-like circuits) that is separated from the memory circuit which data it caches. Typically, such a cache has its own data path and address space, and so is unable or very restricted in its ability to also operate as another independent storage or memory circuit. However, as illustrated in FIG. 1(a), the SRAM arrays provided on companion chip 102 share data-paths formed in hybrid bonds 111 and an address space with the quasi-volatile memory circuits of memory chip 101. Under such an arrangement, even when operating as a cache for the quasi-volatile memory circuit in memory chip 101 (i.e., being mapped into the quasi-memory circuit address space), the SRAM arrays may still serve as a fast-access memory circuit accessible from the separate SRAM address space discussed above. Furthermore, the cache and the fast-access memory

operations can take place over shared data paths. As discussed above, access by host processor **103** is available for both cache access and fast memory access over input and output interface circuits **109** (e.g., an industry-standard DDR5 interface or a high-bandwidth memory (HBM) interface).

In one embodiment, the high-bandwidth internal data buses for data transfers between memory chip **101** and companion chip **102** may also be used for transferring data in a massively parallel fashion between SRAM arrays in companion chip **102**. This facility is particularly advantageous for in-memory computation operations. These internal buses deliver large amounts of data per execution cycle to the high-speed logic, RISC processors, math co-processors, or arithmetic circuit modules on companion chip **102**, without involving moving data over input and output interface **109**. Such an arrangement allows host processor **103** to set up arithmetic or logic operations to be carried out by the logic or arithmetic circuit modules on companion chip **102**, without the data having to move over input and output interface **109**, thereby circumventing the proverbial “von Neuman bottleneck.”

In one embodiment, the SRAM arrays in companion chip **102** are used as cache memory for the quasi-volatile memory circuits only in a one-to-one correlated cache mode (i.e., the addressable unit of storage, such as “page,” is identical in both the quasi-memory array as in the SRAM arrays). However, such an approach may not be ideal for some applications. For example, an SRAM array in companion chip **102** may be configured to be address on a “page” basis, which may be 2 Kbytes, as in some embodiments discussed above. In some operating system software, a page may be defined to be 512 bytes or 1K bytes. As another example, under one industry standard, an addressable data unit based on the width of an industry standard memory interface bus (e.g., 128-bit) may be preferable. In one embodiment, a portion of an SRAM array may be configured to be addressed on a “page-by-page” basis, with the page size configurable, or any suitable addressable data unit to accommodate the requirements of host processor **103**, an operating system, or any suitable application program. The addressing scheme may be fixed, or configurable by software, firmware, or based on host command at “run” time (i.e., dynamically) by setting configuration registers in companion chip **102**, for example.

Because of the number of high-bandwidth internal data buses that are available, parallel multiple-bank (whether concurrent or non-concurrent) operations are possible. While large amounts of data are delivered for arithmetic and logic operations by the high-speed arithmetic or logic circuit modules on companion chip **102**, the next set of data may be fetched in parallel from the quasi-volatile memory circuits in memory chip **101** to be loaded into SRAM arrays in companion chip **102**. Organizing the SRAM arrays and the logic and arithmetic circuit modules in rows and columns, parallel computation tasks (e.g., those used in AI applications) may be various segments of the bank basis (e.g., less than all logical tiles at a time), on a tile column basis or on multiple banks at a time. This operation of the SRAM array may be controlled or allocated by firmware or circuitry (e.g., state machines) on companion chip **102** or by a command set issued by host processor **103**.

In one embodiment, a bank of SRAM arrays may be organized into a tile array of 256 rows by 16 columns, such that a 256-bit internal data bus is associated with one column of the SRAM tiles. In that configuration, 16 parallel 256-bit arithmetic or logic operations may be carried out simulta-

neously for data associated with each bank. Furthermore, in one embodiment, the 16 columns may be divided into four bank segments, for example, such that the 16 parallel operations are 4 sets of different operations, each set corresponding to a bank segment. The SRAM arrays on companion chip **102** may also be organized as bank groups, with each bank group having multiple banks. Independent and parallel operations may be carried out on a bank-group basis. In this manner, the SRAM arrays in the memory chipset of the present invention can be easily allocated in many possible configurations to simultaneously carry out both cache operations and in-memory computation operations.

FIG. 5(b) shows additional variations **533** and **534** that can be implemented for type **521** circuit modules of FIG. 5(a) at surface **212** of substrate **211** of companion chip **102**, according to one embodiment of the present invention.

Some or all of the SRAM arrays **541** may be replaced by arrays of eDRAM, MRAM, phase-change memory, resistive random-access memory, conductive bridging random-access memory or ferro-electric resistive random-access memory, or any suitable combination of these circuits. Some of these memory arrays may provide comparable results in other embodiments of the present invention.

FIG. 6 generally illustrates memory module **600**, according to one embodiment of the present invention, which may be provided in the format of a dual-in-line memory module (DIMM). As shown in FIG. 6, memory module **600** includes controller circuit **602** and memory chipsets **601-0** to **601-15**, each of which may be a memory chip bonded to a companion chip (e.g., integrated circuit **102** described above). Memory module **600** may be mechanically attached to a printed circuit board on which electrical connection are provided (e.g., over an industry-standard data bus) to host computing system **603**. Host computing system **603** may be any computing system, e.g., servers and mobile device, or any other suitable computing device (e.g., any telecommunication switch, router or gene sequencer). While FIG. 6 shows 16 memory chipsets, this number of memory chipsets is merely illustrative and is not intended to be limiting of the present invention. Memory module **600** may include memory chipsets of quasi-volatile memory circuits, in some embodiments, the chipsets may include both quasi-volatile memory circuits and non-volatile circuits, and circuits of another memory type (e.g. DRAM). The specific memory configuration may be optimized to accommodate the expected workloads and power requirements of host system **603**. Controller circuit **602** may be provided as a separate integrated circuit. Controller **602** may be a conventional memory controller or may be specific to operations of chipsets with quasi-volatile memory circuits with on-chipset compute or mathematical operation functions.

According to one embodiment of the present invention, FIG. 7 illustrates integrated circuit **160**, which includes non-memory chip **161** and a memory chipset (e.g., chipset **120** above, which includes memory chip **101** and companion chip **102**). Non-memory chip **161** may include one or more CPUs, GPUs, FPGAs, image sensors, baseband and other signal processors, ethernet and other data communication circuits, or any other suitable logic circuits. In integrated circuit **160**, memory chipset **120** and non-memory chip **161** may be bonded together, signals between memory chipset and non-memory chip **161** electrically connected using, for example, through-silicon vias (TSVs), which improve signal communication speeds and reduce latency between memory chipset **120** and non-memory chip **161** during operation. Another embodiment may use another conventional interconnect, bond or bump technology. For example, memory

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chipset **120** and the non-memory chip **161** may be configured to use any suitable interface technique (e.g., DDR, HBM, or register-to-register data transfer techniques). An interface that implements a register-to-register data transfer protocol may optimize software or hardware performance (e.g., software of an operating system or application executing on a host computer system, or packet switching circuits in a telecommunication router).

According to another embodiment of the present invention, as shown in FIG. 8, integrated circuit **800** includes memory chipset **120** and non-memory chip **161** interconnected by a silicon interposer, exemplified by silicon interposer **801**. Silicon interposer **801** serve as a silicon substrate that provides interconnection conductors, in a manner similar to a printed circuit board. Silicon interposer **801** may provide electrical connections to additional memory chipsets and additional non-memory chips. Silicon interposer **801** provides the advantage of fast signal communication between the interconnected chips, while avoiding packaging challenges, such as heat dissipation.

FIG. 9 is a schematic representation of computing system **900**, which may be a subsystem within a larger host system (e.g., host system **603** of FIG. 6), for example. Computing system **900** may perform specialized applications (e.g. gene sequencing, telecommunication, or automotive and internet of things (IoT) applications). Computing system **900** illustrates that companion chip **102** may be customized and optimized to meet the workloads generated by software application **903**, operating system **902**, and firmware **901** of host processor **103**. In computing system **900**, SRAM arrays **107** or other buffer-type or cache-type memory circuits inside memory chipset **120** that are associated with quasi-volatile or non-volatile memory arrays **110** of memory chip **101** may be managed and configured outside memory chipset **120**. Management optimization may be achieved, for example, by machine learning or digital signal processing techniques.

FIG. 10 is a schematic representation of memory chipset **125**, which is provided battery **1001** or a capacitor on companion chip **102**. Memory chipset **125** is advantageous for applications in which companion chip **102** stores system information (e.g., memory management information, including locations of bad blocks, lookup tables and registers). Memory chipset avoids loss of data when memory chipset **125** loses power. Battery **1001** retains data in any SRAM arrays or other volatile memory circuits on companion chip **102** or memory chip **101**. In the event of a power loss, battery **1001**, firmware on companion chip **102** and dedicated quasi-volatile or other non-volatile backup memory on memory chip **101** allow memory chipset **125** to write such system information (e.g., memory management information) into a non-volatile memory circuit. The stored system information may be recovered at the next power-up.

One advantage of the SRAM arrays on companion chip **102** is power conservation. The standard DDR5 for memory modules permits suspension of refresh operations, when the host system (e.g., host system **603** of FIG. 6) is idle. Some embodiments of the present invention allowing shutting down selected quasi-volatile memory blocks. When refresh suspension is permitted, a user may transfer critical data (e.g., firmware for the memory chipset or meta-data about the up-to-date status of the memory tiles) from the quasi-volatile memory circuits to the SRAM arrays, so that refresh operations on the quasi-volatile memory circuits may be suspended to conserve power. When power resumes, normal operations may be quickly restarted by the firmware in the SRAM arrays of the companion chip. Alternatively, refresh

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operations may be stopped for all quasi-memory circuits, except a selected few. Critical information for resumption of operations (e.g., the firmware for memory chipset) may be stored in the selected few blocks for which refresh operations are maintained.

Integrated Circuit **120** of the present invention may support a paging scheme in a virtual memory system, according to the present invention. FIG. 11 schematically illustrates a paging system using the fast memory circuits (e.g., SRAM circuits) and the quasi-volatile memory circuits of integrated circuit **120**, in accordance with one embodiment of the present invention. Under the paging scheme of one embodiment, companion chip **102** keeps a suitable number of blocks of SRAM circuits **1101** (under a suitable block size, such as 1 byte, 64 bits, 128 bits, 2 Kbytes or any suitable addressable unit), based on the requirements of the intended application or operating system, to service the next incoming read or write command for data at specific locations associated with quasi-volatile memory circuits **1157** in memory chip **101** from host processor **103**.

In FIG. 11, flow chart **1103** is provided to illustrate the operation of this paging system. Initially, at step **1151**, a number of blocks of SRAM circuits **1101** ("memory blocks") are allocated. The blocks of SRAM circuits may be managed or allocated for this purpose using a page table and a suitable data structure, such as "heap," "stack," "list," or any other suitable data structure, as is known to those of ordinary skill in the art. To improve performance, as seen from host processor **103**'s perspective, a memory operation control circuit (e.g., a state machine-based control circuit) in data-path and control circuit **108** of companion chip **102** (see FIG. 1) may be provided. Recall that the actual write operation to quasi-volatile memory circuit **1157** may require up to, for example, 100 nanoseconds, even though the data may be read out from a copy stored in SRAM circuits **1101** over a very short time (e.g., 10 nanoseconds). Accordingly, companion chip **102** avoids stalling service to host processor **103** by scheduling the slower write operations to the quasi-volatile memory circuits **1157** in the background. In particular, a memory block holding data to be written in quasi-volatile memory circuit **1157** must be allowed to finish the write operation of its entire content into quasi-volatile memory circuits **1157**. This requires having sufficient number of memory blocks available to service a suitable number of next incoming read or write commands from host system **103**.

At step **1152**, the memory operation control circuit determines the number of memory blocks that have not been allocated and, at step **1153**, determines if the number of unallocated memory blocks exceeds a threshold. If so, at step **1154**, there is sufficient unallocated memory blocks remaining without requiring a currently allocated memory block to write back its content to quasi-volatile memory **1157** to make room. Otherwise, at step **1155**, a currently allocated memory block is selected based on an "eviction" policy and its data "evicted" or written back into the corresponding locations in quasi-volatile memory circuits **1157** in memory chip **101**. A suitable eviction policy may be, for example, the "least recently accessed" (i.e., the block among all allocated blocks that has not been read for the longest time). At step **1156**, the data in the selected memory block is written back to the corresponding locations (as identified in the page tables) back to quasi-volatile memory circuits **1157**. During this time, the memory operation control circuit monitors the "ready or busy" state of the applicable quasi volatile memory bank and when the bank is not busy, companion chip **102** deems the write operation com-

plete and returns to step 1152. As there are sufficient unallocated memory blocks to handle the read and write access requests from host processor 103, while a number of incomplete write operations back to quasi-volatile memory 1157 may be proceeding in parallel, read and write requests from host processor 103 would not be stalled for an incomplete write operation.

The method represented by flow chart 1103 is applicable to and is advantageous for cache operations too. Of course, in a cache application, there is usually no need to select which memory block to write back.

While the above detailed description provides as HNOR memory string arrays (e.g., those described in Non-provisional Application II) as a primary example of quasi-volatile and non-volatile memory circuits on the memory chip. Other types of quasi-volatile and non-volatile memory circuits (e.g., the VNOR memory string arrays, described in Non-provisional Application III) also may be used in various embodiments of the present invention and achieves the advantages discussed above. For example, hybrid bonding allows the VNOR memory arrays the high-bandwidth interconnections to the SRAM arrays and the computation logic elements in the companion chip (e.g., SRAM circuits 541 and arithmetic and logic circuits 544 on companion chip 102 of FIGS. 5a and 5b). Whether HNOR memory string arrays or VNOR memory string arrays are used to provide quasi-volatile and non-volatile memory circuits, sense amplifiers and other high-performance, low-voltage logic circuitry may be implemented on the companion chip and electrically connected through the hybrid bonds to provide data, to take advantage of having the data from the sense amplifiers being in close proximity to both the SRAM circuits and the computation logic circuits.

The above detailed description is provided to illustrate specific embodiments of the present invention and is not intended to be limiting. Numerous variations and modifications of the present invention are possible. For example, in this detailed description and in the drawings, SRAM circuits are mentioned or used extensively to illustrate the present invention. However, the present invention is applicable to other fast memory circuits as well. The use of SRAM circuits to illustrate fast memory circuits herein is not intended to be limiting. The present invention is set forth in the accompanying claims.

The invention claimed is:

1. An integrated circuit, comprising:

a first semiconductor die having memory circuits configured into a plurality of first memory modules formed above a substrate layer in the first semiconductor die, each first memory module comprising a plurality of memory cells organized into one or more arrays of memory strings, with the memory strings of each array being provided along first dimension and a second dimension, such that each memory string in the array extends along a third dimension, the first dimension, the second dimension and the third dimension being each mutually orthogonal to both the other two dimensions; and

a second semiconductor die having memory circuits configured into a plurality of second memory modules, wherein each second memory module is configurable to be associated with a group of one or more first memory modules, such that the memory circuits within each second memory module and memory circuits of its associated first memory modules are interconnected by

interconnections formed by wafer-level or chip-level bonding between the first and second semiconductor dies.

2. The integrated circuit of claim 1, wherein the wafer-level or chip-level bonding comprises one of: hybrid bonding, direct interconnection bonding, and micro-bump bonding.

3. The integrated circuit of claim 1, wherein memory circuits in the second memory modules have a lower latency than memory circuits of the first memory modules.

4. The integrated circuit of claim 1, wherein the memory circuits within the first memory modules comprise quasi-volatile memory circuits or non-volatile memory circuits and the memory circuits with the second memory modules comprise one or more of: static random-access memory (SRAM) circuits, dynamic random-access memory (DRAM) circuits, embedded DRAM (eDRAM) circuits, magnetic random-access memory (MRAM) circuits, embedded MRAM (eMRAM) circuits, spin-transfer torque MRAM (ST-MRAM) circuits, phase-change memory (PCM), resistive random-access memory (RRAM), conductive bridging random-access memory (CBRAM), ferro-electric resistive random-access memory (FRAM), carbon nanotube and memory.

5. The integrated circuit of claim 4 wherein signals communicated between memory circuits within a first memory module and memory circuits within its associated second memory modules are multiplexed and demultiplexed to share the interconnections.

6. The integrated circuit of claim 1, wherein the second semiconductor die is fabricated under a manufacturing process optimized for fabricating CMOS logic circuits.

7. The integrated circuit of claim 1, wherein the second semiconductor die further comprising sense amplifiers, registers or data latches, and logic circuits formed thereon.

8. The integrated circuit of claim 1, further comprising on the second semiconductor die a plurality of internal data buses each providing read and write accesses to a corresponding group of second memory modules.

9. The integrated circuit of claim 8, further comprising arithmetic and logic circuits formed on the second semiconductor die, wherein the arithmetic and logic circuits are organized into logic modules and wherein the logic modules access over the internal buses the memory circuits in their corresponding groups of second memory modules.

10. The integrated circuit of claim 9, wherein signals from memory circuits in each first memory module are configurable to be connected by the bonds to a corresponding one of a plurality of data output circuits in the second semiconductor die, such that the corresponding data output circuit provides the signals to the second memory module associated with the first memory module.

11. The integrated circuit of claim 10, wherein each data output circuit comprises sense amplifiers.

12. The integrated circuit of claim 10, wherein each data output circuit is associated with one or more multiplexer circuits in the first semiconductor die, the multiplexer circuits selecting data signals from the first memory module to be sent to its corresponding data output circuit.

13. The integrated circuit of claim 10, further comprising master-slave registers, provided between the data output circuits and the second memory modules, wherein a slave latch of each master-slave register holds a current data output of one of the data output circuits, while a master latch of each master-slave register receives a next data output from one of the data output circuits.

14. The integrated circuit of claim 10, wherein each logic module of the arithmetic and logic circuits is associated with one or more second memory modules, and wherein each logic module accesses its associated second memory modules over the internal data buses corresponding to the associated second memory modules.

15. The integrated circuit of claim 10, wherein each second memory module comprises a plurality of memory cells, a plurality of word lines and a plurality of bit lines for selecting the memory cells within the second memory module for read or write access, and wherein the second memory module is configurable to perform as a bit-wise multiplier that receives, as a first operand, a data value represented by a selected set of word lines and, as a second operand, a data value represented by a selected set of bit lines, with a product value resulting from the bit-wise multiplier stored into the memory cells to selected memory cells.

16. The integrated circuit of claim 15, wherein a plurality of the logic modules are provided to sum the product values written into the selected memory cells.

17. The integrated circuit of claim 10, wherein each logic module in the second semiconductor die is configured as one of: an adder circuit, a divider circuit, a Boolean operator circuit, a multiplier circuit, a subtractor circuit, a RISC processor, a math co-processor, and a multiplexer circuit.

18. The integrated circuit of claim 10, further comprising an on-chip control circuit in the second semiconductor die that controls operations of both the second memory modules and the logic modules.

19. The integrated circuit of claim 18, wherein the on-chip control circuit implements caching or paging of data from the memory circuits of the first memory modules in the memory circuits of the second memory modules.

20. The integrated circuit of claim 19, wherein the caching or paging of data is carried out using a block size determined by a page size fixed in the memory circuits of the first memory modules.

21. The integrated circuit of claim 20, wherein the caching or paging of data is carried out using a programmable block size.

22. The integrated circuit of claim 21, wherein the programmable block size is one of a plurality of block sizes each corresponding to units of data transfer in one or more industry-standard memory interface protocols.

23. The integrated circuit of claim 21, wherein the programmable block size is set by a host processor.

24. The integrated circuit of claim 19, wherein the on-chip control circuit allocates the memory circuits in blocks, maintaining at least a predetermined number of unallocated blocks.

25. The integrated circuit of claim 24, wherein when the number of unallocated blocks falls below the predetermined number, the on-chip control circuit selects one of the allocated blocks for data transfer to the memory circuits of the first memory modules.

26. The integrated circuit of claim 18, further comprising a memory interface formed in the second semiconductor die accessible by a host processor and data paths that connect among the first memory modules, the second memory modules and the logic modules and the memory interface.

27. The integrated circuit of claim 26, wherein the host processor accesses the first memory modules and the second memory modules over the memory interface using an access scheme that is based on mapping the first memory modules

and the second memory modules, respectively, to first and second non-overlapping portions of addresses in an address space.

28. The integrated circuit of claim 27, wherein hot data or metadata are stored and read from memory at addresses in the second portion of the address space.

29. The integrated circuit of claim 28, wherein the host processor requests data from a location in the memory circuits in the first semiconductor die by presenting an address for the location as mapped to the address space, and wherein the on-chip control circuit provides, as the requested data, data retrieved from the second memory module.

30. The integrated circuit of claim 29, wherein the second memory module from which the data is retrieved is allocated to serve as cache memory for the first memory module to which the address is mapped.

31. The integrated circuit of claim 30, wherein the first memory modules are organized as memory banks, and wherein the second memory modules serving as cache memories for first memory modules of a memory bank in the first semiconductor die form a corresponding memory bank in the second semiconductor die.

32. The integrated circuit of claim 31, wherein a plurality of memory banks form a bank group, the bank group further comprises a first data bus accessible by each of the memory banks within the bank group.

33. The integrated circuit of claim 32, wherein the integrated circuit comprises a plurality of the bank groups, wherein the integrated circuit further comprising a second data bus accessible by each of the bank groups, and wherein each of the bank groups accesses the second data bus through the first data bus of its own bank group.

34. The integrated circuit of claim 1, wherein each memory circuit of the second memory modules is configured to be one of: a dual-ported memory circuit and a single-ported memory circuit.

35. The integrated circuit of claim 1, wherein each memory string comprises a VNOR memory string.

36. The integrated circuit of claim 1, wherein each memory string comprises a HNOR memory string.

37. The integrated circuit of claim 1, further comprising support circuitry formed under the memory arrays and at a substrate layer of the first semiconductor die.

38. The integrated circuit of claim 37, wherein the first and second semiconductor dies are bonded in a "flip-chip" or "face-to-face" orientation.

39. The integrated circuit of claim 37, wherein the substrate layer of the first semiconductor die comprises an insulator.

40. The integrated circuit of claim 37, wherein through-silicon vias (TSV) are provided to connect the support circuitry to circuitry formed on the second semiconductor die.

41. The integrated circuit of claim 37, wherein the support circuitry comprises voltage sources for reading, programming or erase.

42. The integrated circuit of claim 41, wherein the support circuitry further comprises logic circuits operating on voltages lesser in magnitude than voltages of the voltage sources for programming and erase.

43. The integrated circuit of claim 42, wherein the support circuitry further comprises selection transistors for routing signals from memory circuits of the first memory modules to the interconnections.

44. The integrated circuit of claim 42, wherein the support circuit further comprises word line-related control circuits.

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45. The integrated circuit of claim 44, wherein the word line-related control circuits comprise word-line decoders.

46. The integrated circuit of claim 37, wherein sense amplifiers for sensing memory circuits of the first memory modules are formed at a planar surface of the second semiconductor die and are connected to memory circuits in the first memory modules over the interconnections.

47. The integrated circuit of claim 42, wherein each signal generated by the voltage sources for reading, programming and erase is routed by conductors that are shielded by virtue of being provided between grounded conductors.

48. The integrated circuit of claim 1, further comprising vertical thin-film transistors that route signals from memory circuits within each first memory module to the interconnections.

49. The integrated circuit of claim 1, further comprising non-memory circuits formed on a third semiconductor die, wherein the third semiconductor die is bonded to the second semiconductor die to allow the non-memory circuits and circuitry on the second semiconductor die to be electrically interconnected.

50. The integrated circuit of claim 49, wherein through-silicon vias are formed on one or both of the second semiconductor die and the third semiconductor die to provide signal paths.

51. The integrated circuit of claim 49, where the non-memory circuits are part of a computing system.

52. The integrated circuit of claim 51, wherein the computing system comprises one or more of: central processing units, graphics processing unit, memory controllers, RISC processors, math co-processors, servers, mobile devices, telecommunication switches, routers and gene sequencers.

53. The integrated circuit of claim 52, wherein the non-memory circuits are part of a computing system.

54. The integrated circuit of claim 53, wherein the computing system comprises one or more of: central processing units, graphics processing unit, memory controllers, RISC processors, math co-processors, servers, mobile devices, telecommunication switches, routers and gene sequencers.

55. The integrated circuit of claim 1, further comprising non-memory circuits formed on a third semiconductor die, wherein the second and third semiconductor dies are both bonded to a silicon interposer substrate to allow the non-memory circuits and circuitry on the second semiconductor die to be electrically interconnected by conductors formed in the silicon interposer substrate.

56. The integrated circuit of claim 1, further comprising a battery that supplies backup power to the integrated circuit.

57. The integrated circuit of claim 1, wherein the substrate layer in each of the semiconductor dies is divided into a plurality of tiles, with each tile in the first semiconductor die being associated with one of the first memory modules, and with each tile in the second semiconductor die being associated with one of the second memory modules.

58. The integrated circuit of claim 57, wherein each tile in the first semiconductor die is mapped one-to-one onto a corresponding tile in the second semiconductor die, and wherein one or more signals from each tile in the first semiconductor die are provided to circuitry at the substrate layer of the corresponding tile by the interconnections or by thru-silicon vias.

59. The integrated circuit of claim 58, wherein the one-to-one mapping maps a tile in the first semiconductor die to the tile in the second semiconductor die that is closest in proximity.

60. The integrated circuit of claim 57, wherein each tile in the first semiconductor die comprises a fast memory circuit

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section, in which the fast memory circuits are placed, and one or more data output sections, in which data output circuits are placed, provided on one or both sides of the fast memory circuit section.

61. The integrated circuit of claim 60, wherein the second semiconductor die has further formed thereon a plurality of logic module each comprising a plurality of logic circuits, and wherein each logic module is associated with one of the tiles.

62. The integrated circuit of claim 61, wherein the tiles are organized into one or more computing units, with each computing unit comprising:

a first plurality of contiguous rows of the tiles in the second semiconductor die each associated with a group of the second memory modules;

a second plurality of contiguous rows of the tiles in the second semiconductor die each associated with a group of the logic modules, and

a plurality of computing data buses each facilitating data transfer among the memory circuits in the first group of second memory modules and the logic circuits in the group of logic module.

63. The integrated circuit of claim 62, wherein the each computing unit is configurable to implement a neural network having neurons each parameterized by a matrix of parameter values stored in the second memory modules of the computing unit, wherein input data to the neural network is stored as vectors in the first memory modules associated with those second memory modules, and wherein the logic modules implement a matrix multiplications between matrices of the parameter values and the vectors.

64. The integrated circuit of claim 62, wherein a group of the logic modules form one of: a central processing unit (CPU) core, a graphics processing unit (GPU) core, field-programmable gate arrays (FPGAs), and an embedded controller.

65. The integrated circuit of claim 57, wherein the tiles are arranged on each semiconductor die in a formation of rows and columns, and wherein an internal bus on the second semiconductor die is provided for read and write access to second memory modules associated with a column of tiles.

66. A computing system comprising a plurality of the integrated circuits, each integrated circuit being as provided in the integrated circuit of claim 57, the integrated circuits being interconnected by a memory interface bus, wherein a subset of the integrated circuits are configured to operate in a pipelined manner.

67. The computing system of claim 66, wherein a first subset of integrated circuits are configured such that the memory circuits in the first memory modules comprise NOR memory strings that perform a logical function.

68. The computing system of claim 67, wherein the first subset of integrated circuits implements one or more content-addressable memory circuits.

69. The computing system of claim 67, wherein members of the first subset of the integrated circuits are further configured to perform a parallel search function over content addressable memory circuits.

70. The computing system of claim 69, wherein a second subset of integrated circuits are configured to perform data-intensive computations.

71. The computing system of claim 70, wherein the computing system is configured to transfer results of the data-intensive computations to be used in the parallel search function over the content addressable memory circuits.

72. An integrated memory module having an input and output interface, comprising:

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a plurality of the integrated circuits, wherein one of the integrated circuits is provided by the integrated circuit of claim 1; and

a controller managing the integrated circuits so as to provide access to the integrated circuits over the input and output interface.

73. The integrated memory module of claim 72, wherein the memory module conforms to a dual-inline memory module format.

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